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Study of Machining Processes Using Finite Element Analysis

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1. INTRODUCTION

1.1 Subject and objective of the thesis

Toothed wheels are the most common mechanical components used for motion and power transmission. The Power skiving method is a modern gear machining technology, which combines characteristics of Gear shaping and Gear hobbing. This process presents certain advantages over traditional methods.

The purpose of this thesis is the analysis of the mechanical behaviors of the cutting tool as well as the chip of the workpiece, emphasizing the study of stresses, cutting forces, and deformations resulting from the Power skiving process. The analysis will be done through the Finite Element Method (FEM), using the Ansys software.

Furthermore, an extensive reference will be made to the various types of gears, to certain matching methods, as well as a comparative evaluation between them.

1.2 Structure of thesis

Chapter 1 presents the objective and structure of the thesis.

Chapter 2 will review the literature regarding the types, geometric characteristics, and materials of gears, as well as machining methods.

Chapter 3 describes the numerical model and the finite element method used for the analysis.

Chapter 4 presents the process of studying stresses and deformations in the simulation program environment using finite elements.

Chapter 5 contains the results of the process for the different cutting positions.

2. LITERATURE REVIEW

2.1 Categories of toothed wheels

2.1.1 Spur gears

Also known as straight-tooth frontal gears, they constitute one of the most widespread types of gears. Their teeth are parallel to the axis of rotation, thus allowing direct engagement with the corresponding gear. Their design is simple and they exhibit relatively low losses in terms of power transmission.

Some of their main disadvantages are increased noise as well as the large dynamic loads they create. For this reason, their use is not recommended for heavy loads.

Some of the main characteristics for their design and geometry are the module (m), which is defined as the ratio of the pitch circle diameter (d) to the number of teeth (N), the pressure angle (ϕ), whose typical value has been set at 20°, and the tooth width (b), which affects the gear's stability and load transmission capacity.

Due to the above factors, spur gears are mainly used in applications where efficiency is a priority and noise is not a deterrent factor, such as in industrial equipment or even robotic systems.

Figure 2.1: Spur gears

2.1.2 Helical gears

In this type of gear, the teeth are at an angle relative to the wheel axis. Due to this geometry, transmission occurs more smoothly, with reduced vibrations and noise, in contrast to the case of spur gears. Furthermore, they are ideal for high-speed applications, and due to the large contact surface area, they have the ability to transmit greater power with less wear.

At the same time, they have disadvantages in terms of design because their construction requires greater precision, thus increasing production costs. Also, due to the inclined arrangement of the teeth, axial loading is created. Therefore, good support of the shafts and bearings is required to maintain system performance.

A very important technical characteristic is the helix angle of the teeth, with a typical value ranging from 15° to 30°. It is the angle formed between the tooth surface and the gear's axis of rotation. Angles of 15°-20° are used in applications with high precision and lower axial load, while 20°-30° are used in cases where high power is required. There are two types of angles, right-hand and left-hand helix angles.

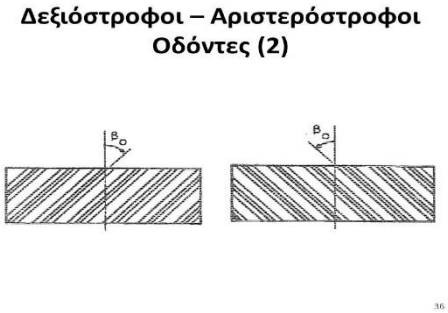


Figure 2.2 Right-hand and Left-hand Teeth

Additional parameters that should be considered during the design process include the maximum strength of the gear teeth, the surface finish, and the dimensions of the tooth profiles, which affect the generated noise and vibrations. Finally, prior to production, the power and torque transmission capacity must be calculated.

Figure 2.3 Helical gear

2.1.3 Herringbone gears

Essentially, they consist of two gears with teeth of opposite inclination joined in one body. Due to the two opposite inclination angles, axial forces are canceled out. Furthermore, continuous contact between the two teeth is maintained, thanks to which less noise is produced than with spur or helical gears. Herringbone gears have frequent industrial application because they demonstrate great strength under high loads.

Their construction is quite complex and has a higher cost compared to other types of gears. This is mainly because the two sections of the gear must be perfectly aligned; otherwise, wear due to use will be greater and noise will increase. Additionally, special machinery is often required for the construction of large-sized herringbone gears, and construction can take up to several weeks.

Figure 2.4 Herringbone gear

2.1.4 Bevel gears

Bevel gears have a conical shape and are used to transmit motion between two intersecting shafts.

Depending on the type of teeth, they are classified into straight-tooth and helical-tooth gears. In the first case, manufacturing is simpler; however, their load-carrying capacity is lower and the generated noise is higher due to the instantaneous contact between the teeth. In contrast, gears with helical teeth produce less noise because the contact between the teeth occurs gradually. They can also operate at higher speeds, although they generate axial forces.

The larger the pitch diameter, that is, the diameter at which the teeth come into contact, the greater the torque that can be transmitted.

An important technical characteristic is the angle formed between the gear and its axis of rotation, which determines how two gears will mesh with each other. In addition, in helical bevel gears the helix angle affects the axial forces; as the helix angle increases, the operation of the system becomes smoother.

Due to their geometry, bevel gears allow motion transmission between shafts at various ratios, which is why they are widely used in gearboxes and vehicle differentials.

Figure 2.5 Bevel gears

2.1.5 Hypoid gears

Hypoid gears are a variation of bevel gears, but with the main difference that the gear axes do not intersect, thus creating an eccentricity due to which higher torque transmission is achieved. They also usually have spiral teeth, which contribute to withstanding heavy loads.

High-strength materials such as alloy steels are commonly used for their manufacture, as they can withstand wear as well as the high temperatures generated by heavy loads. To enhance their durability, gears are typically subjected to electrical discharge machining and various heat treatment processes during production.

Based on these characteristics, their disadvantages include the complex and costly manufacturing process, the need for regular wear inspection, and their sensitivity to overheating, which requires frequent lubrication.

They are often used in vehicle gearboxes, wind turbines, differentials, and generally in systems where high precision and resistance to high speeds and loads are required.

Figure 2.6 Hypoid gears

2.1.6 Internal gears

These are gears in which the teeth are located on the inner side of the ring wheel. They usually cooperate with spur gears in planetary transmission systems.

Such a system usually consists of an internal gear, one or more planet gears, and a small central sun gear that engages with the planet gears.

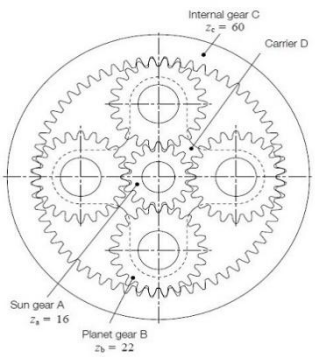


Figure 2.7 Planetary gear system

They have a fairly compact design; due to the internal contact of the gears, vibrations and noise are reduced. Furthermore, the way the teeth engage leads to fewer friction losses, and they are ideal for applications with high loads.

Their construction is difficult and requires methods such as broaching, EDM, and gear skiving, which increase production costs. Some manufacturers have somehow modified some of the above processes to reduce time and consequently cost.

Some common applications of internal gears are internal gear pumps, CNC machines, planetary gearboxes, and hydraulic systems.

Figure 2.8 Internal gear

2.1.7 Gear Rack and Pinion

The gear rack and pinion pair is a mechanism used mainly for converting rotary motion into linear motion. It consists of a spur gear (Pinion) which engages with a rectangular-section toothed bar (Rack), whose direction of motion depends on the direction of rotation of the wheel .

In other cases, the gear rotates on the shaft that remains fixed, or the gear has a fixed shaft and the rack performs translational motion.

A characteristic example of use is in vehicle steering systems, where when the driver turns the steering wheel, the gear rotates, moving the bar linearly. This displacement is transferred to the wheels, and in this way, the vehicle changes direction.

A similar motion conversion system is that of the screw and nut. In comparison, the Gear rack and Pinion exhibits less friction, thus it is more efficient and performs better in high-speed applications. On the other hand, a

significant comparative disadvantage is the backlash. For this reason, the screw-nut system is preferred in cases where high precision is required.

Figure 2.9 Gear Rack and Pinion

2.1.8 Worm gear

The system consists of a gear (worm wheel) and a shaft with a spiral form (worm) which rotates, transmitting motion to the toothed wheel. The two axes are perpendicular to each other.

The toothed wheel cannot drive the worm, and at the same time, the worm can only rotate in one direction, providing the property of self-locking without additional mechanisms. Furthermore, this system offers high speed reduction and quiet operation. Due to these characteristics, it is mainly used in lifting machinery, power transmission systems, and elevators.

However, at high transmission ratios, it exhibits a high friction coefficient, leading to low efficiency, overheating, and wear. To reduce friction, materials with coatings are often used. Efficiency can range from 50% to 90%.

For better performance, most manufacturers use specially designed threads that reduce contact between the two surfaces, consequently reducing friction.

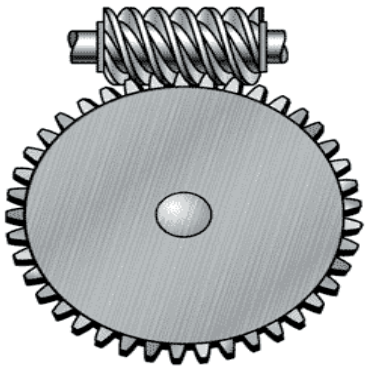


Figure 2.10 Worm gear

2.1.9 Screw gears

This is a separate category of gears. They are essentially helical gears that transmit motion between axes at a 90-degree inclination, in contrast to a previous category (Bevel gears) where the gear axes were parallel .

Compared to other motion transmission mechanisms for perpendicular axes, such as Worm gears, they have lower friction and thus better efficiency because the tooth contact is smoother. They also produce less noise and exhibit less wear, thus increasing their service life.

Some of the most common machining processes for Screw gears are Milling, Cold forging, and laser cutting

For their construction, steel is mainly used due to its high and long-lasting resistance to stresses. Additionally, other materials such as aluminum, bronze, plastics, and other composite materials are used depending on the requirements of each application.

A key technical characteristic is the gear ratio, which is determined by the numerical ratio of the teeth and the helix angle. The gear ratio affects the speed and torque of the system; if the ratio is less than 1, then the output speed will be greater than the input speed. While if it is greater than 1, the torque on the output shaft will increase, consequently reducing the speed.

Figure 2.11 Screw gears

2.2. Basic terminology of toothed wheels

2.2.1 Number of teeth (z)

It refers to the number of teeth a gear has and is one of the most critical parameters in the construction of a gear wheel as it directly affects the gear ratio as well as the strength of the gear, which increases proportionally with (z).

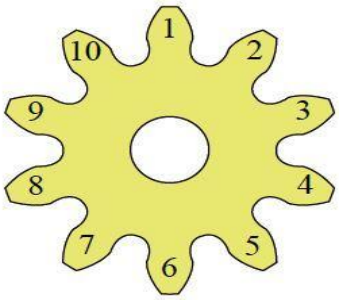


Figure 2.12 Gear with number of teeth z = 10

2.2.2 Tooth hight (h)

The total height of a tooth results from the sum of the addendum (ha) and the dedendum (hf).

$$h = h_a + h_f$$

- Addendum is defined as the distance from the pitch circle to the tip of the tooth.
- Dedendum is defined as the distance from the pitch circle to the root of the tooth.

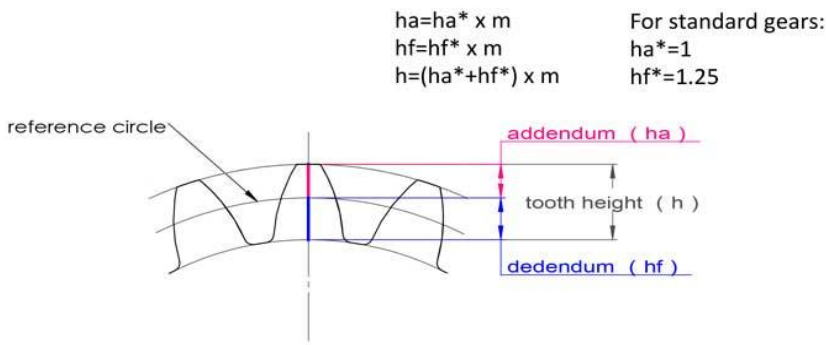


Figure 2.13 Calculation of tooth height

2.2.3 Module (m)

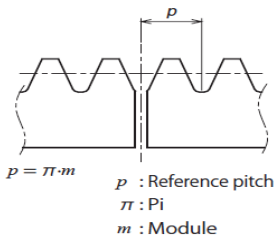
Module is the unit that determines the size of the teeth of a gear wheel. It is one of the most important factors as it affects the gear ratio and compatibility with other gears.

The basic formula is :

$$m = D_p / z$$

- D_p = Pitch Diameter (mm)
- z = Number of teeth

Furthermore, the Pitch equals the product of the module and π, so the larger the module, the greater the distance between the teeth.



Unit: mm

Module	Pitch (p)	Tolerance
m0.5	1.57	−0.05 −0.15
m0.8	2.51	−0.05 −0.25
m1	3.14	−0.1 −0.3
m1.5	4.71	
m2	6.28	
m2.5	7.85	−0.1 −0.4
m3	9.42	
m4	12.57	
m5	15.71	
m6	18.85	
m8	25.13	
m10	31.42	

Figure 2.14 Module–Pitch Relationship Example

2.2.4 Gear ratio (i)

The gear ratio is defined as the ratio of the revolutions of the driven gear to the revolutions of the driving gear.

$$i = n_2 / n_1$$

Essentially, the gear ratio indicates how many more or fewer revolutions the driven gear has compared to the driving gear.

2.2.5 Pressure angle (α)

The pressure angle is the angle that the contact force of the gear teeth forms with the perpendicular at the contact point. In the past, its typical value was 14.5°. Now, the value most commonly encountered is 20° and 25° for heavier loads.

The smaller the pressure angle, the smoother and noiseless the operation, and the load-carrying capacity decreases. While the larger it is, the more noise and wear increase, but also the load-carrying capacity.

The formula for calculating the pressure angle differs depending on the category and geometry of each gear wheel.

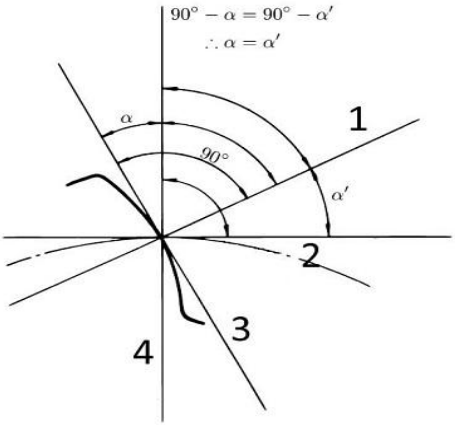


Figure 2.15 Example of pressure angle

2.2.6 Diameter

To determine the size of a gear wheel, we need the reference diameter (d).

$$d = z \cdot m$$

Two other important dimensions are the tip diameter (da), i.e., the diameter of the imaginary circle passing through the tips of the teeth, and the diameter of the circle passing through the base of the teeth, called the root diameter (df).

$$d_a = d + 2p$$

$$d_f = d - 2.5p$$

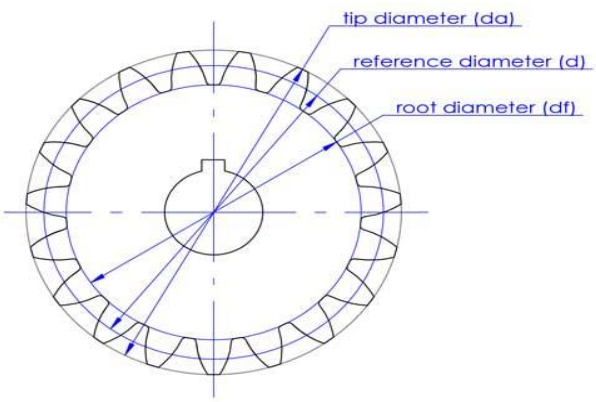


Figure 2.16 Visualization of gear diameters

2.2.7 Clearance

Clearance is defined as the perpendicular distance from the tip of one tooth to the root of a tooth of the mating gear wheel.

$$c = 0.25 \cdot m$$

0.25 has been set as a constant factor in most gear systems, ensuring that there is sufficient space between the teeth of the two wheels; however, in some cases, it can be adjusted due to operating conditions.

Furthermore, another important parameter is the center distance (a), which measures the distance between the centers of the two engaging gear wheels.

$$\alpha = d_1 + d_2 / 2$$

where d_1 and d_2 are the diameters of the two gears.

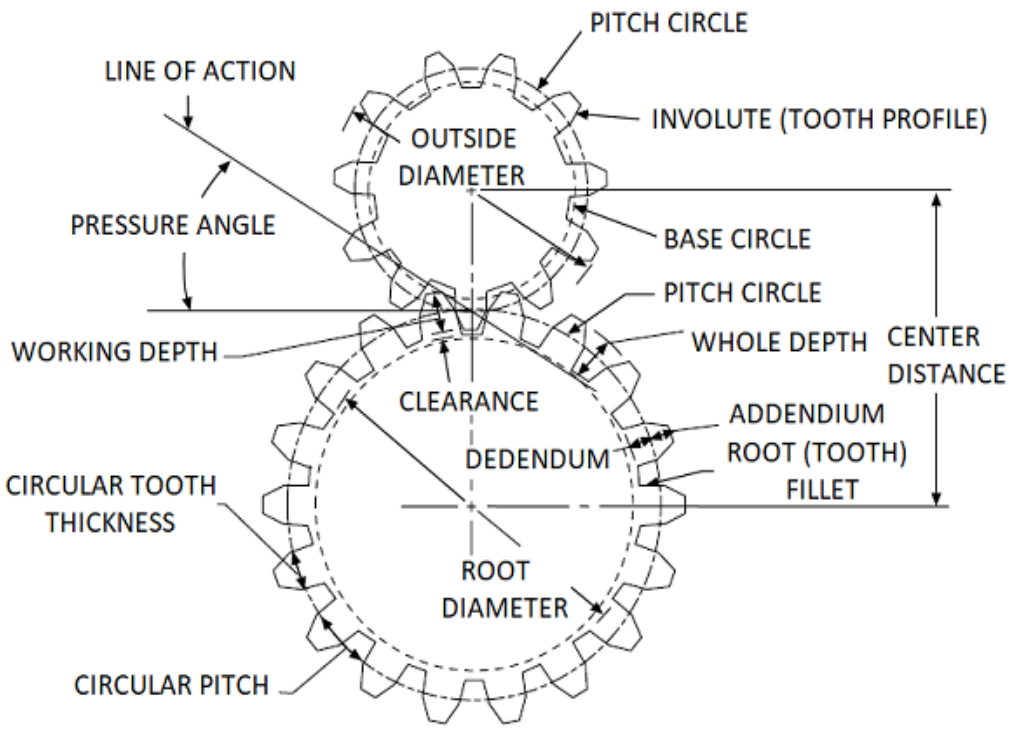


Figure 2.17 Example of clearance and center distance

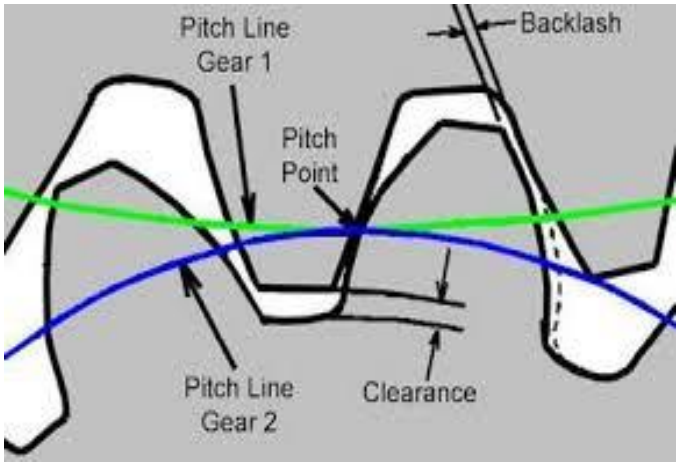


Figure 2.18 Gear engagement

As can be seen from Figure 2.18, Backlash is another important characteristic as it defines the gap that exists between the teeth of two mating gears, which is necessary for the smooth operation of the system..

2.2.8 Tooth profile

In industry, most gears used have teeth based on the involute curve. This profile allows for smooth and efficient motion transfer, maintaining a constant rate of rotation between the cooperating wheels. Furthermore, this specific geometry is particularly practical in design and manufacturing, as it allows for easy production of gears with a different number of teeth, as long as they have the same module.

Some of the characteristics that define gears with an involute tooth profile are:

- Simplified manufacturing process
- Low production cost
- Durable construction
- Smooth operation
- Design flexibility

The involute curve results from tracing the path that a point on a cord follows as it is unwound from a circle. This circle, known as the base circle, is decisive for the tooth profile, and its diameter is calculated based on the pressure angle and the pitch circle of the gear.

The formula for the base circle:

$$d_b = d \cdot \cos(\alpha)$$

where:

- d_b = diameter of the base circle
- d = diameter of the pitch circle
- α = pressure angle

This specific geometry is extremely efficient, as it allows the teeth to transmit motion with minimal losses and without sudden changes in force. Furthermore, the pressure angle of the gears affects the interaction of the teeth, resulting in smoother force transmission.

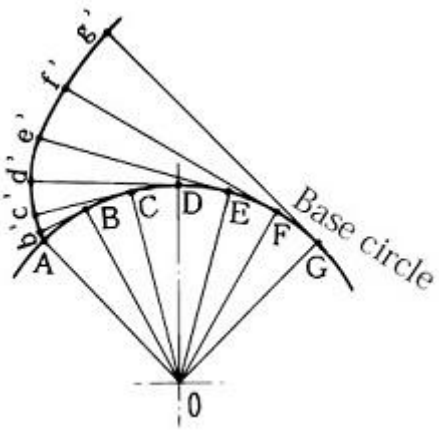


Figure 2.19 Involute curve

2.2.9 Gear engagement

Gear engagement occurs at a specific point, which lies on the pitch circle of each gear. This point is where the teeth of the wheels begin to come into contact and transfer motion from one gear to the other. The common tangent of the two base circles, which determine the dimensions of the gears, is referred to as the line of contact or line of action. This line determines the path along which the contact point (P1 - P2 - P3) moves as the teeth interact with each other.

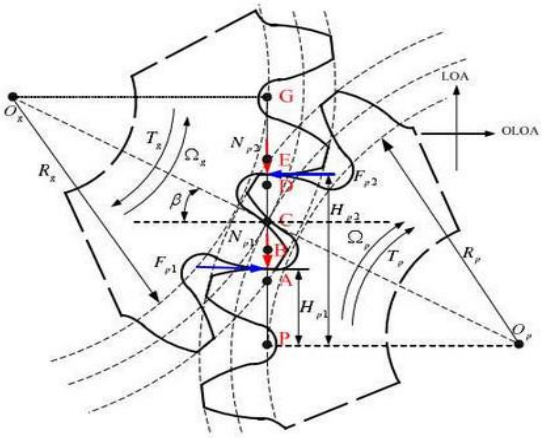


Figure 2.20 Gear engagement

2.3 Gear materials

Gear materials play an important role in the lifespan and performance of gears. The choice of each material depends on many factors such as wear, loading, temperature resistance, and differs from application to application. Below, reference will be made to the materials most frequently encountered int the construction of gear wheels.

Carbon Steels (π.χ. C45, S45C)

Carbon steels are among the most used materials, a fact due to their high mechanical strength and their ability to undergo various thermal treatments to enhance surface hardness.

Stainless Steels (π.χ. SUS303, SUS304)

They are used more often in applications that must resist corrosion and do not require regular maintenance. However, they have low mechanical strength.

Alloy Steels (π.χ. SCM440, SCM415)

They exhibit high resistance to wear and heat; for this reason, they are ideal for high-stress applications. Like carbon steels, they can undergo thermal treatments such as carburizing to enhance their surface hardness.

Non-Ferrous Materials (Aluminum, Copper)

They are characterized by wear resistance, low weight, and have a low friction coefficient. Their production cost is expensive because casting processes are required for their manufacture. These materials are usually used for gears such as worms and helical gears.

Plastic Materials (π.χ. PA, POM)

Plastic gears are often encountered in applications requiring low friction and minimal lubrication, making them ideal for use in environments where lubricants are not allowed, such as in food processing equipment. They are manufactured mainly through injection molding, a method that allows mass production at low cost and reduced manufacturing time.

Cast Copper Alloys (π.χ. CAC502, CAC702)

Although copper alloys are expensive and their production is difficult, due to their excellent wear resistance, they are used for the construction of specialized gears such as worms.

2.4 Thermal and Chemical treatments

Quenching and Tempering

It is the process of heating a metal and then rapidly cooling it to increase its hardness. This is followed by tempering to improve fatigue resistance. Quenching is primarily applied to steel, and the type of quenching varies depending on the cooling method. After quenching, the steel becomes harder but also more brittle; therefore, it must undergo tempering to reduce brittleness and restore some of the toughness lost during the initial thermal treatment.

Case Hardening

It is a treatment applied to gears to increase their surface hardness while keeping the core tough and resistant to impacts.

The three main methods of case hardening are as follows:

1. Induction Hardening

Initially, it uses electromagnetic induction to heat the gear's surface, followed by rapid cooling. With this method, we achieve selective hardening on specific points of the gear wheel with precise temperature control and in a short time.

2. Flame Hardening

We use a flame to heat the gear's surface, followed by rapid cooling. This process has a low cost, but unlike induction hardening, it is not very well controlled and can cause overheating or even deformation in certain areas of the gear.

3.Nitriding

This is a chemical surface treatment aimed at case hardening. During nitriding, nitrogen or carbon is diffused into the gear's surface at lower temperatures compared to previous methods (500-600°C).

Carburizing

Initially, the metal is heated to a temperature of about 850-950°C. Then, the gear is exposed to a carbon-rich gas or powder. Carbon penetrates the surface layers of the metal. Finally, cooling occurs, by immersing the gear in a liquid (e.g., water, oil), thus causing carbon recombination and the creation of a hard surface layer.

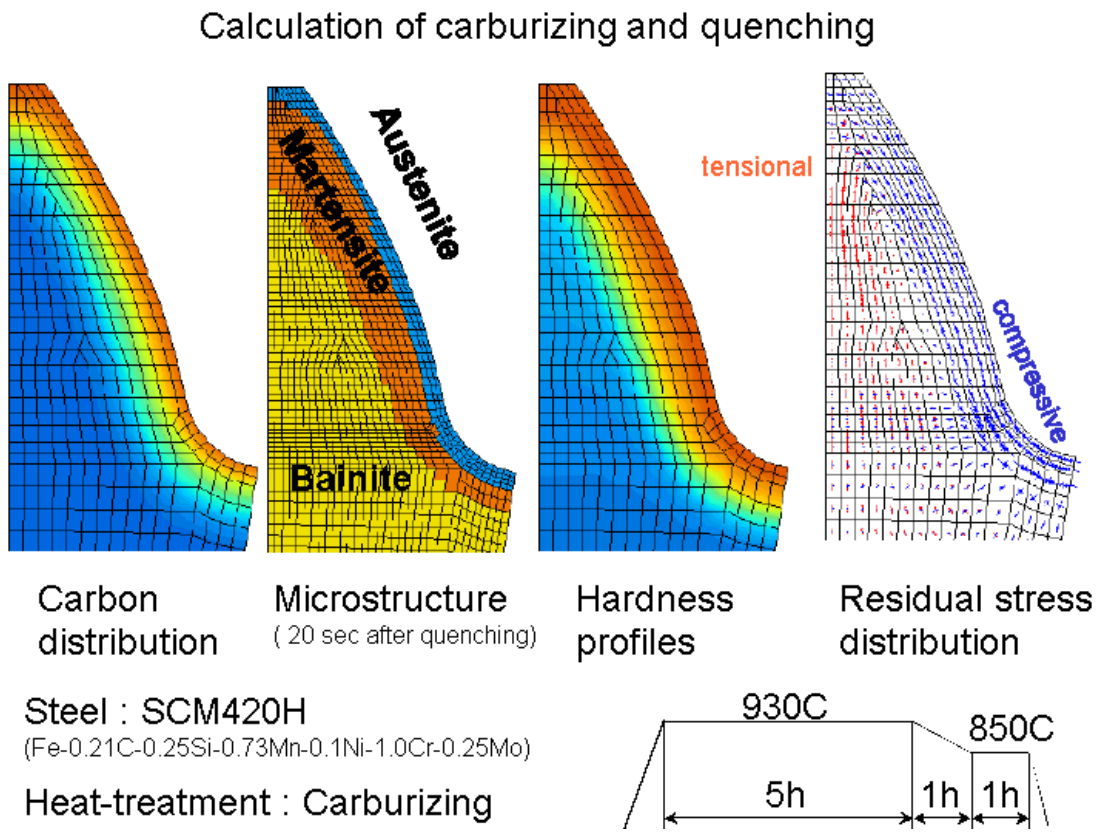


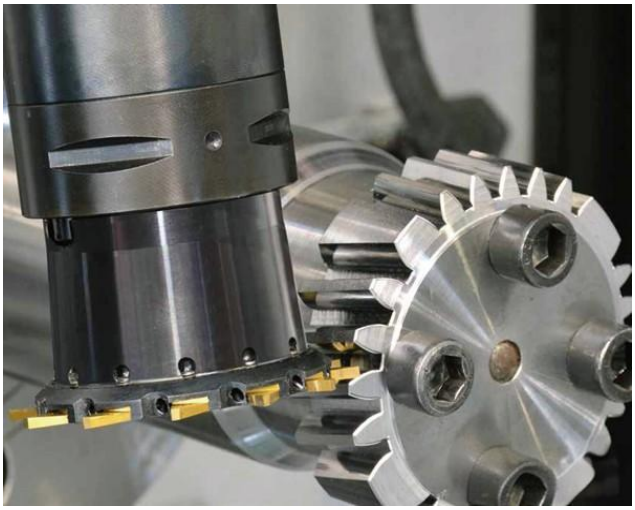
Figure 2.21 The effect of carburizing on the mechanical properties of the tooth

2.5 Gear Manufacturing Technologies

There are various gear manufacturing technologies, with the choice depending on factors such as required accuracy, production cost, and quantity. Each method has its own advantages and limitations, which determine its selection according to the needs of the application. Next, we will examine some of the main methods used for gear manufacturing.

2.5.1 Gear Milling

This is one of the oldest methods and is now mainly used for the manufacturing of straight and helical gears. However, all types of gears can be produced using this technique. The process employs specialized milling cutters with a predetermined profile and module, adapted to the type of gear being manufactured. During machining, the circumference of the gear is divided into as many segments as there are teeth. The cutting tool then penetrates the gear and removes material, creating the groove between the teeth. The shape of the milling cutter determines the tooth profile..



Σχήμα 2.22 Gear Milling

Once the cutting of one groove is completed, the wheel is rotated by a predetermined angle, so that the tool repeats the process for the next tooth. During cutting, the workpiece remains stationary, while the tool moves axially to remove the material. This process, known as Index Milling, is repeated until all teeth are completed.

2.5.2 Gear hobbing

This is one of the most common gear production techniques. During machining, the workpiece performs a rotational motion around its axis, while the hob simultaneously performs both rotational and translational motion in a direction parallel to the axis of the workpiece.

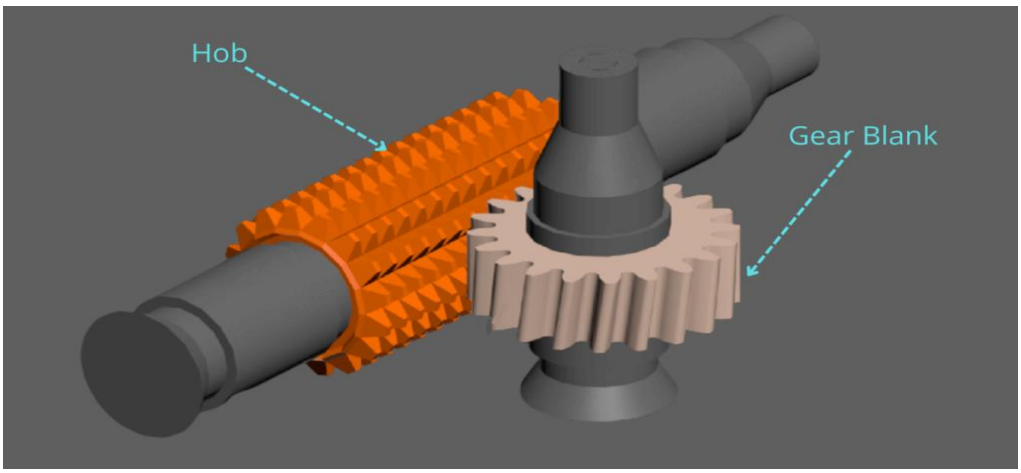


Figure 2.23 Gear Hobbing

Gear hobbing belongs to the generating methods, i.e., unlike other machining techniques, the final tooth profile is not the exact imprint of the cutting tool but results from the combined motion of the tool and the workpiece.

The cutting tool (hob) has a helical shape and carries multiple cutting edges, each designed for efficient material removal and smooth tooth formation.

An important parameter in the hob's geometry is the existence of a clearance angle on the back of the cutting teeth, which reduces friction and improves chip flow. Furthermore, this design facilitates tool maintenance, as it allows grinding of the cutting edges without altering their original shape.

The number of teeth on the tool, its diameter, and the helix angle directly affect cutting performance and quality. A larger number of cutting teeth means more rolling positions per revolution, which improves the smoothness of the final tooth profile.

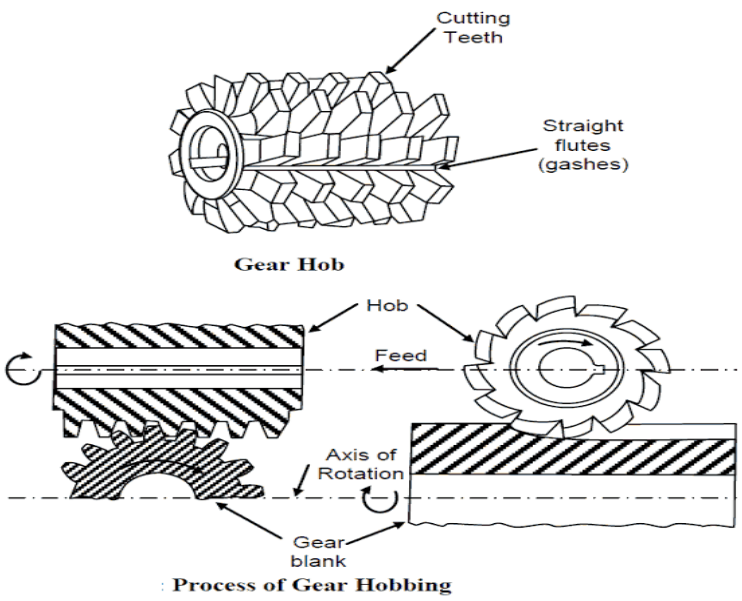


Figure 2.24 Cutting tool geometry

2.5.3 Gear planing

Like Gear hobbing, this belongs to generating technologies. In this method, the specimen rotates around its axis, and the cutting tool consists of a gear rack that moveso reciprocally in a direction perpendicular to the axis of the workpiece.

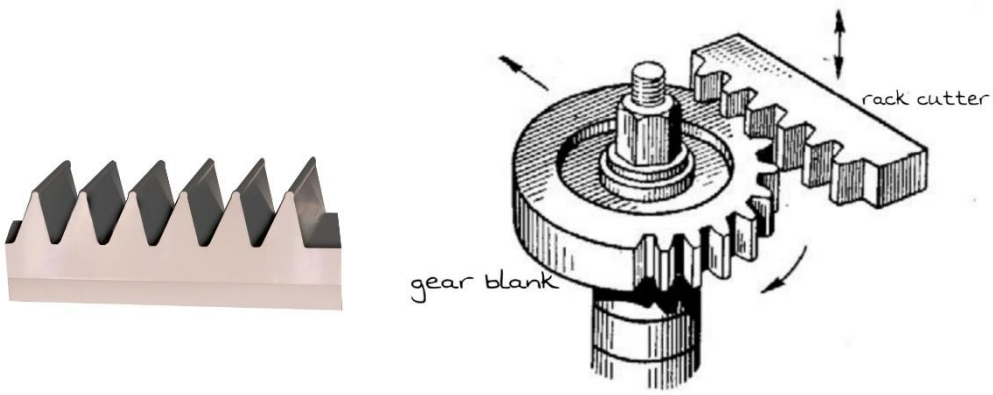


Figure 2.25 Gear rack and simulation of the Gear planing process

The cutting tool removes material only in one direction (downwards). After completing the reciprocating motion of the hob, the gear rotates by a small predetermined step, and in this way, the remaining teeth are formed.

Gear planing is a time-consuming, high-precision process with the capability of machining external and internal gears. It is mainly used for large, heavy-duty gears.

2.5.4 Gear shaping

Gear shaping also belongs to the generating methods. The cutting tool is cylindrical and has external cutting teeth. The cutting tool performs a reciprocating motion (up-down); furthermore, both the tool and the workpiece rotate around their axes. This process continues with the tool gradually moving inward until the desired depth of cut is achieved.

Gear shaping is used for manufacturing internal and external gears.

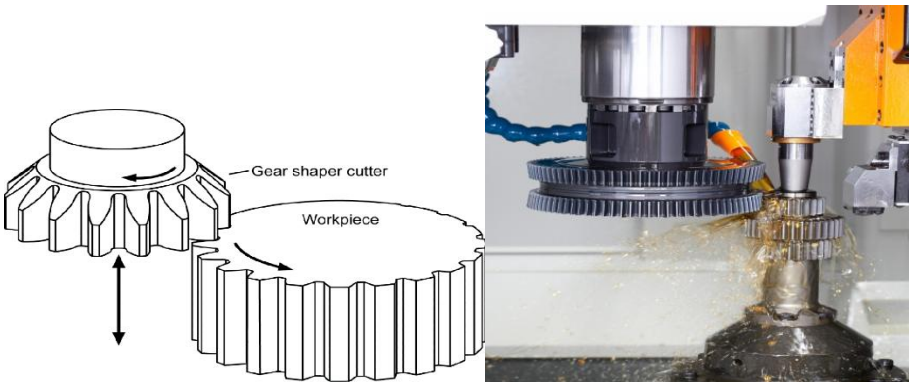


Figure 2.26 Gear shaping

he shaper cutter has a form similar to a gear and is mainly made of High-Speed Steel (HSS), Carbide, and CBN. Below, I list the basic types of cutting tools for this method.

- Disk-type cutter: The most common form for this process; it has a common form with a gear and is used for external gears.

- Hub-type cutter: It has a similar form with a shorter profile and a hole in the center; it is mainly used for small or medium-sized internal gears.

- Pinion-type cutter: It has a small size with a form similar to a pinion. Due to its size, it has better access to narrow recesses and has greater wear resistance. Suitable for high-precision applications in internal and external gears.

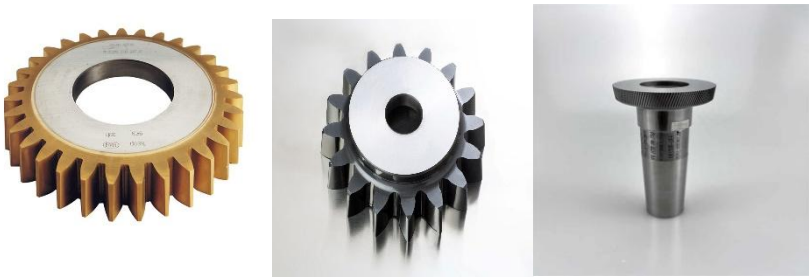


Figure 2.27 Disk-type cutter, Hub-type cutter, and Pinion-type cutter, from left to right.

2.5.5 Power skiving

Power Skiving is a modern gear machining method that combines the advantages of hobbing and shaping methods, offering faster and more accurate production.

The process involves the simultaneous rotation of the cutting tool and the workpiece around their axes, at different speeds. The angle of the cutting tool (usually 10° - 45°) relative to the gear allows material removal in a gradual and continuous manner, which reduces machining time.

Advantages of Power Skiving

- Excellent machining speed, compared to other traditional methods.
- Excellent surface quality, reducing the need for additional finishing.
- Ability to machine both internal and external gears, without restriction.
- Flexibility in producing small and large series, eliminating the need for tool changes.
- Reduction in production costs, due to completing the machining in a single stage.

Comparison with other gear machining methods

- Compared to hobbing, Power Skiving is more flexible, as it can also manufacture internal gears.
- Compared to shaping, Power Skiving is faster, as it does not require intermittent motion and can process materials in continuous motion.
- Compared to broaching, Power Skiving is more flexible, as it does not require special tools for each application.

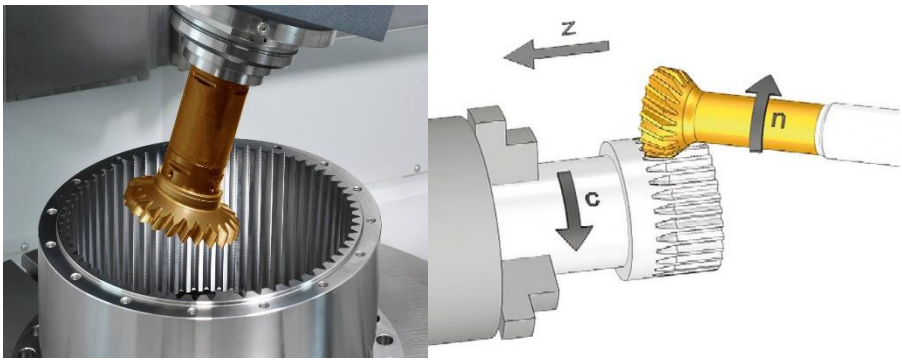


Figure 2.28 Example of Power skiving process

3. Simulation and Analysis Model

3.1 Introduction to the finite element method

The Finite Element Method (FEM) is one of the most widespread numerical techniques for simulating complex mechanical problems. It is widely used for analyzing stresses, deformations, thermal conductivity, and fluid dynamics in mechanical structures and processes.

The physical basis of FEM lies in the discretization of a continuous field (e.g., a mechanical component) into a finite number of smaller elements, which are connected via nodes. In each element, the basic equations of equilibrium, compatibility, and physical laws (such as Hooke's law) are applied to obtain an approximate solution for the overall problem.

3.1.1 Theoretical background and basic equations

The fundamental principle of the method is based on the Differential Form of the Equilibrium Equation for a continuous medium:

$$\nabla \cdot \sigma + f = 0$$

where:

- σ represents the stress tensor
- f is the external forces per unit volume (body forces)

This equation expresses the balance of forces acting on the body.

To calculate stresses and strains in a material, knowledge of the relationships between strains (ε) and displacements (u) is required, as well as the use of Hooke's law:

$$\varepsilon = \tfrac{1}{2} (\nabla u + (\nabla u)^T)$$

$$\sigma = D \cdot \varepsilon$$

where:

- ε is the strain of the material
- ∇u is the gradient of the displacement vector, i.e., how displacement changes in space
- $(\nabla u)^T$ is the transposed gradient, calculates material strains in the reverse direction • D is the elasticity tensor
- D is the elasticity tensor

3.1.2 Division of the body into finite elements

In FEM, the continuous body is divided into a set of elements (triangular, quadrilateral, tetrahedral, or hexahedral elements), and the unknown displacements (u) are calculated only at specific nodes.

Initially, for each element, the equilibrium equation is formed in its discrete form:

$$K\,U = F$$

where:

- K is the stiffness matrix
- U is the displacement vector at the nodes
- F is the external force vector

The stiffness matrix K is derived from the integration of the individual elements based on the equilibrium equations and material conditions.

In ANSYS analysis, the software automatically calculates the stiffness and solves the system of equations, returning values for displacements, stresses, and strains in each element of the model.

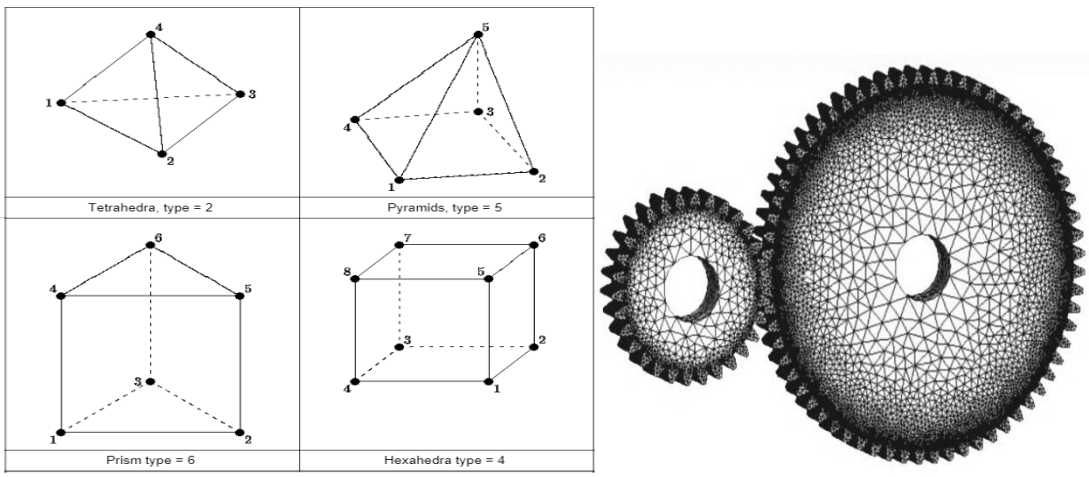


Figure 3.1 Example of nodes and division into elements

3.1.3 Element Analysis and Stiffness Calculation

The stiffness matrix K is the central tool in FEM analysis. To better understand its nature, let's examine its structure in an element, e.g., in a quadrilateral or triangular element.

In FEM methodology, the K for each element is formed by discretizing the differential equilibrium equation, while data related to materials(such as E for Young's modulus and v for Poisson's ratio) are introduced to define the material properties. The integration process over the elements (even using Gauss quadrature for approximate calculations) leads to an accurate computational estimate of the material's dynamic characteristics.

Sometimes, depending on the model, it may be necessary to differentiate for analysis under different loads or loading conditions, for example, if the material is nonlinear or incompressible.

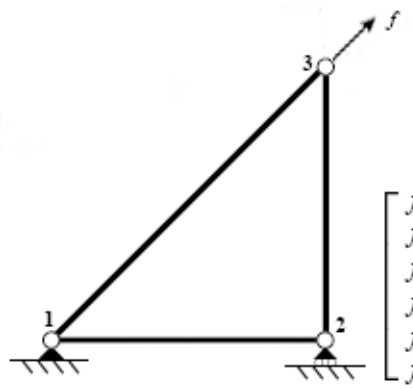
Calculation of the Stiffness Matrix:

If we consider an element with quadrilateral geometry, the stiffness matrix K could be expressed as:

$$K=\int_A B^T D\, B\, dA$$

where:

- **A** is the area of the element
- **B** is the strain-displacement matrix (derivatives of strains with respect to spatial coordinates)
- **D** is the elasticity matrix
- The stiffness matrix is obtained by integrating over the entire element domain



$$\mathbf{f} = \mathbf{K}\mathbf{u}$$

Or

$$\begin{bmatrix} f_{x1} \\ f_{y1} \\ f_{x2} \\ f_{y2} \\ f_{x3} \\ f_{y3} \end{bmatrix} = \begin{bmatrix} k_{11} & k_{12} & k_{13} & k_{14} & k_{15} & k_{16} \\ k_{21} & k_{22} & k_{23} & k_{24} & k_{25} & k_{26} \\ k_{31} & k_{32} & k_{33} & k_{34} & k_{35} & k_{36} \\ k_{41} & k_{42} & k_{43} & k_{44} & k_{45} & k_{46} \\ k_{51} & k_{52} & k_{53} & k_{54} & k_{55} & k_{56} \\ k_{61} & k_{62} & k_{63} & k_{64} & k_{65} & k_{66} \end{bmatrix} \begin{bmatrix} u_{x1} \\ u_{y1} \\ u_{x2} \\ u_{y2} \\ u_{x3} \\ u_{y3} \end{bmatrix}$$

Figure 3.2 Example of stiffness matrix

3.2 Modeling of cutting tool

Extracting the geometry of the cutting tool is a critical stage in simulating the Power Skiving cutting process. This geometry does not arise directly from the profile of the workpiece gear but is shaped considering specific parameters, such as the tool inclination angle and the gear helix angle.

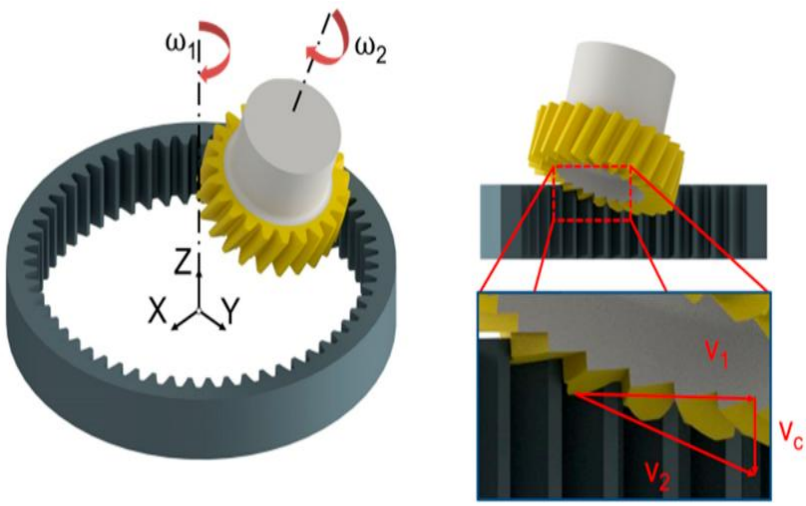


Figure 3.3 Basic kinematics of Power Skiving

3.2.1 Creation of the Cutting Tool Profile

The basic form of the tool results from the reference profile of the workpiece gear, which is intersected at an angle equal to the tool's inclination. This process allows the cutting profile to adapt to the conditions of Power Skiving, considering the kinematic properties of the process. The final tool profile is described by a spline curve, which allows accurate representation of the tool surfaces and improves simulation accuracy.

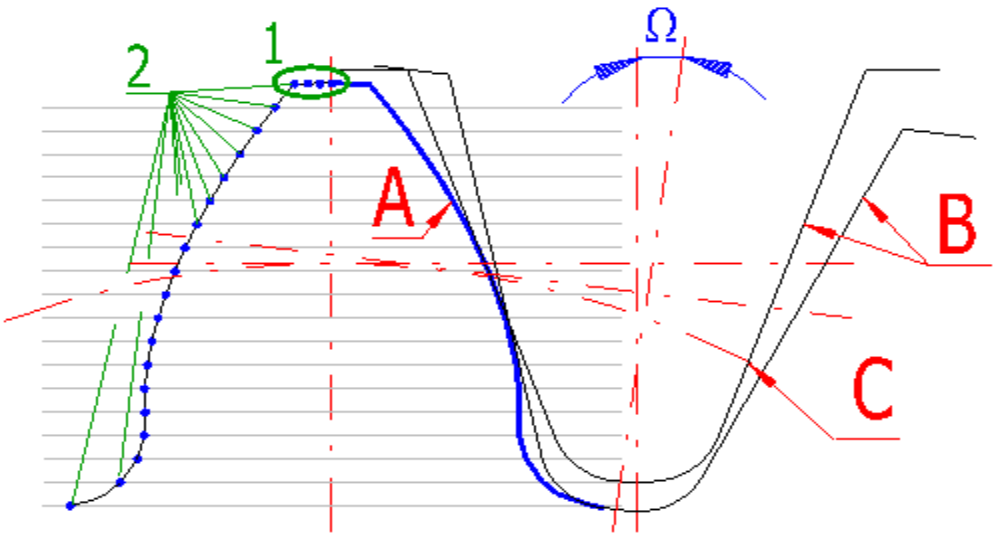


Figure 3.4 Spline design

3.2.2 Integration of Additional Modifications

Depending on the application requirements, additional features can be incorporated into the tool geometry, such as edge rounding and chamfers, which affect the cutting dynamics and the surface quality of the final product. These modifications are introduced through simulation, allowing the user to shape the tool according to their needs.

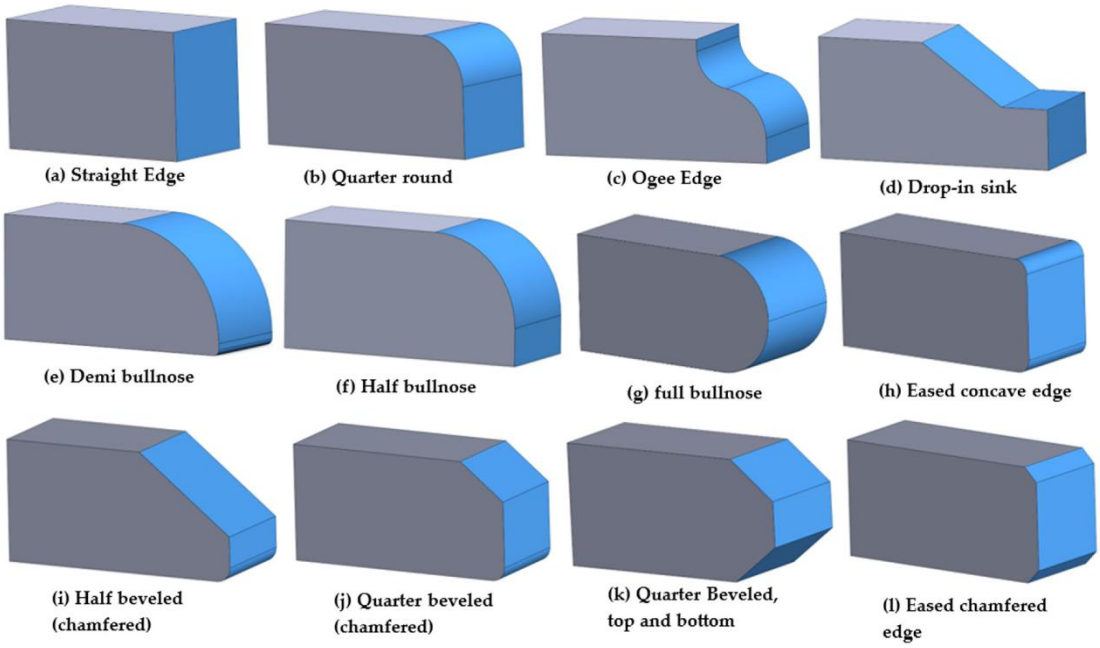


Figure 3.5 Cutting tool edge geometry

3.2.3 Placement of the Tool in Three-Dimensional Space

To represent the cutting process, the tool is placed in successive positions in three-dimensional space, known as Revolving Positions (RP). These positions determine the path followed by the cutting edge during cutting. The primary motion of the process is the rotation of the tool, while the other motions are synchronized with it.

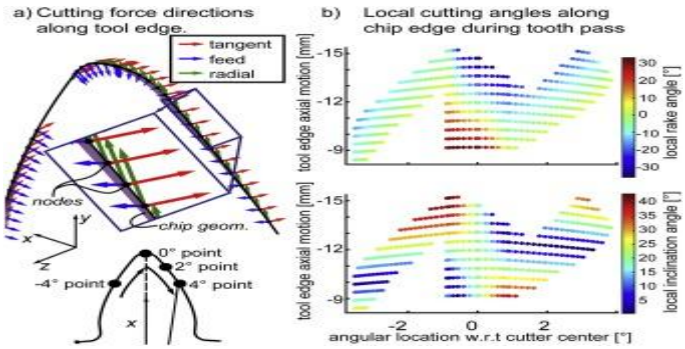


Figure 3.6 Cutting angles in Power skiving

3.2.4 Creation of the Cutting Surface and Determination of the Trajectory

The tool's motion is recorded through successive RPs, creating a surface that describes the cut in space. This surface allows analysis of how the tool removes material from the workpiece. The tool and the workpiece material are combined in the simulation to calculate the volume of material removed in each pass.

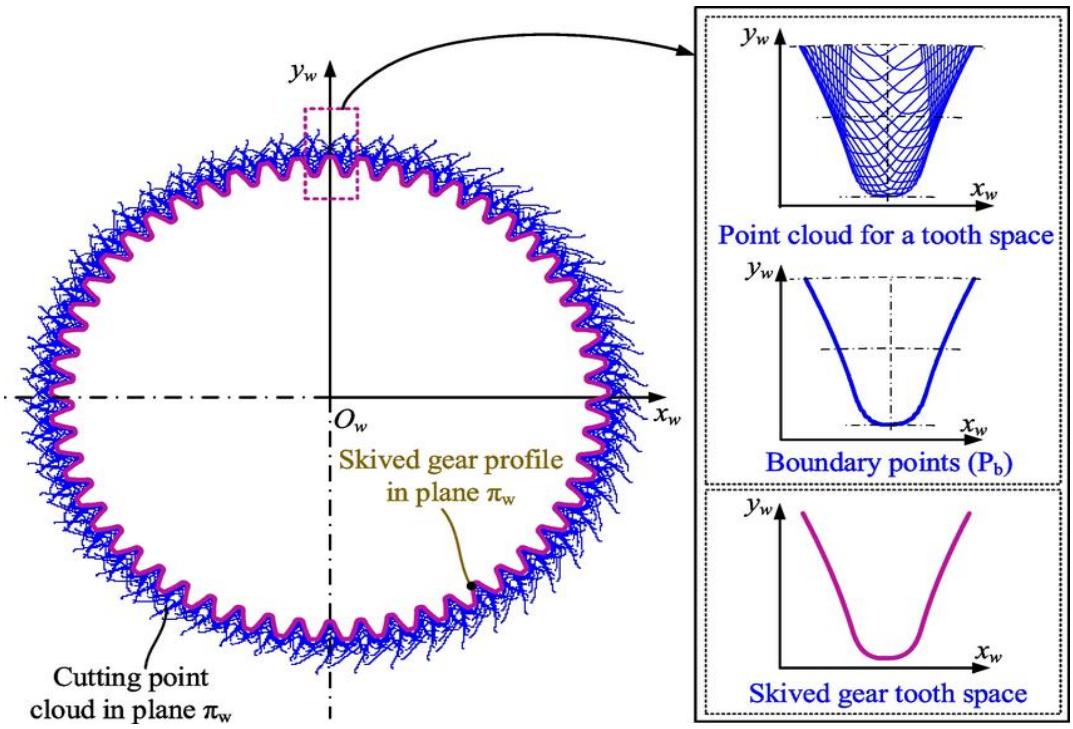


Figure 3.7 Paths traced by the cutting tool during machining

3.3 Analysis of the Solution Process in ANSYS

3.3.1 Definition of the Problem

The simulation in ANSYS concerns the Power Skiving cutting process, aiming to analyze the stresses, cutting forces, and deformations occurring in the cutting tool and the workpiece. The main parameters considered are:

- Geometry of the tool and workpiece
- Material properties (Young's modulus, Poisson's ratio, tensile strength)
- Rotational speed and cutting angle
- Contact forces and friction conditions
- Thermal effects due to the cutting (if required)

3.3.2 Creation of the model in ANSYS

1. Geometry and Materials

The geometry of the cutting tool and workpiece can be imported via CAD models or designed in ANSYS DesignModeler. Material properties are defined from the literature or the software's database.

2. Meshing

Mesh quality affects the accuracy of the results.

Typically used are:

- Tetrahedral or hexahedral elements, depending on the model's geometry.
- Mesh refinement in critical contact areas.

Mesh convergence analysis helps ensure stable results independent of mesh size.

3.3.3 Definition of Conditions and loads

1. Boundary Conditions

- The rotation of the cutting tool is applied as a kinematic constraint.
- The workpiece is constrained in the necessary degrees of freedom.
- Friction conditions are defined using the Coulomb friction model.

2. Application of Forces and Torques

- The torque on the tool is set according to operational requirements.
- Cutting forces are monitored to evaluate stresses.

3.3.4 Selection of Analysis Method and Solution

1. Analysis Type

Depending on the simulation requirements, the following are used:

- Explicit Dynamics: For dynamic phenomena with high cutting speeds.
- Transient Structural: For the temporal evolution of stresses.
- Thermal-Structural Analysis: If examination of thermal effects is required.
- Static structural: For analyzing models under static or nearly static loads.

2. Solution Process

ANSYS uses the Finite Element Method (FEM), applying numerical techniques for model analysis. The solution is based on problem discretization and the application of convergence criteria for solution stability.

3.3.5 Analysis and Visualization of Results (Post-Processing)

After the simulation is completed, the following are extracted and analyzed:

- Von Mises stress distribution on the tool and workpiece
- Cutting forces and system deformations.
- Temperature variations (if thermal analysis is included).
- Diagrams of stress evolution relative to the cutting angle

4. COMPUTATIONAL ANALYSIS

This chapter presents the simulation process in the Ansys software. Through this process, the analysis of stresses and deformations developed during Power skiving is achieved, offering useful information for process optimization. For this specific simulation, different chip geometries have been selected, each corresponding to the position of the tool during machining. The purpose of the thesis is to analyze the stresses and deformations developed in the two bodies during machining as the tool cuts the workpiece and the chip momentarily contacts the tooth surface, exerting pressure.

4.1 Επιλογή προγράμματος προσομοίωσης

ANSYS is one of the most powerful software packages for engineering simulations, offering excellent capabilities for analyzing processes such as Power Skiving. Using advanced algorithms and analysis methods, ANSYS allows accurate modeling of complex physical phenomena, such as cutting forces, thermal interactions, and stresses. The platform offers excellent meshing flexibility, allowing optimization of simulation accuracy according to geometry needs. It also provides powerful tools for more efficient design and tool wear analysis, enabling process improvement and accurate prediction of Power Skiving performance.

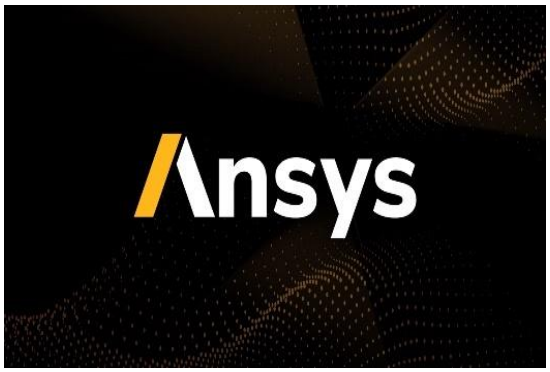


Figure 4.1 Ansys 2025 R2 software

4.2 Geometry creation

For the design of the cutting tool tooth, I used Ansys SpaceClaim due to its compatibility with other tools from the same company and the ease it offers the user.

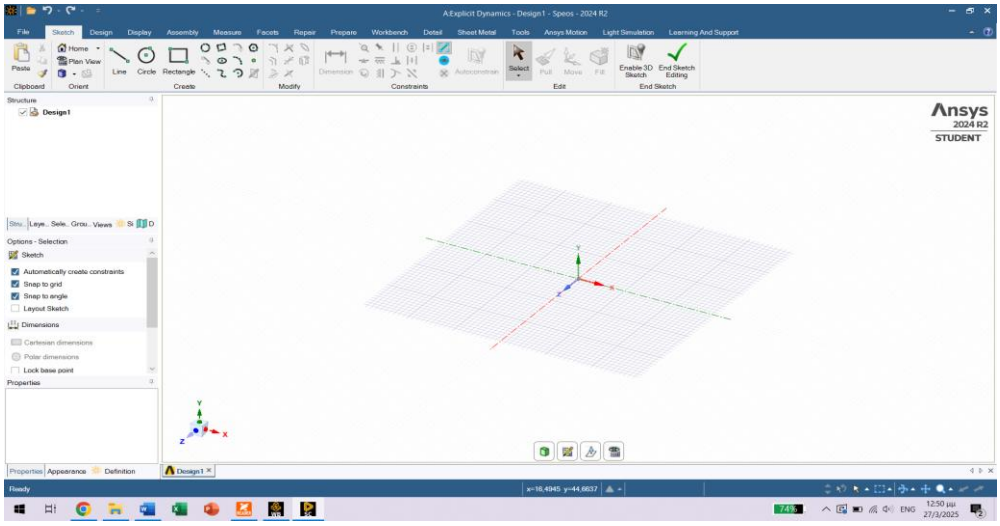


Figure 4.2 Ansys Space claim

According to the methods for selecting the cutting tool geometry, which were analyzed in the previous chapter, I have chosen the geometry I will use for this specific simulation with the following characteristics:

Ιδιότητα	Τιμή
Μήκος X	6,299 mm
Μήκος Y	6,9264 mm
Μήκος Z	10 mm
Όγκος	293,54 mm^3
Μάζα	2,3102e-003 kg
Κέντρο μάζας X	0,36963 mm
Κέντρο μάζας Y	8,3792 mm
Κέντρο μάζας Z	3,6155 mm

Initially, I designed the outline of the tooth as a 2D sketch.

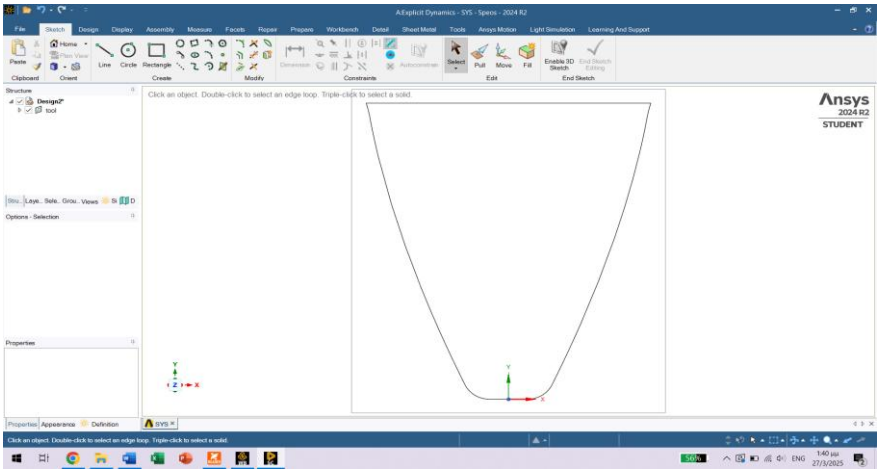


Figure 4.3 2D sketch

Then, using the Fill tool, I fill the interior of the sketch with material. Next, I select Pull to give the desired dimension (10 mm), and finally, I apply a 3° angle to the top side of my geometry to create the angle at which it cuts the workpiece.

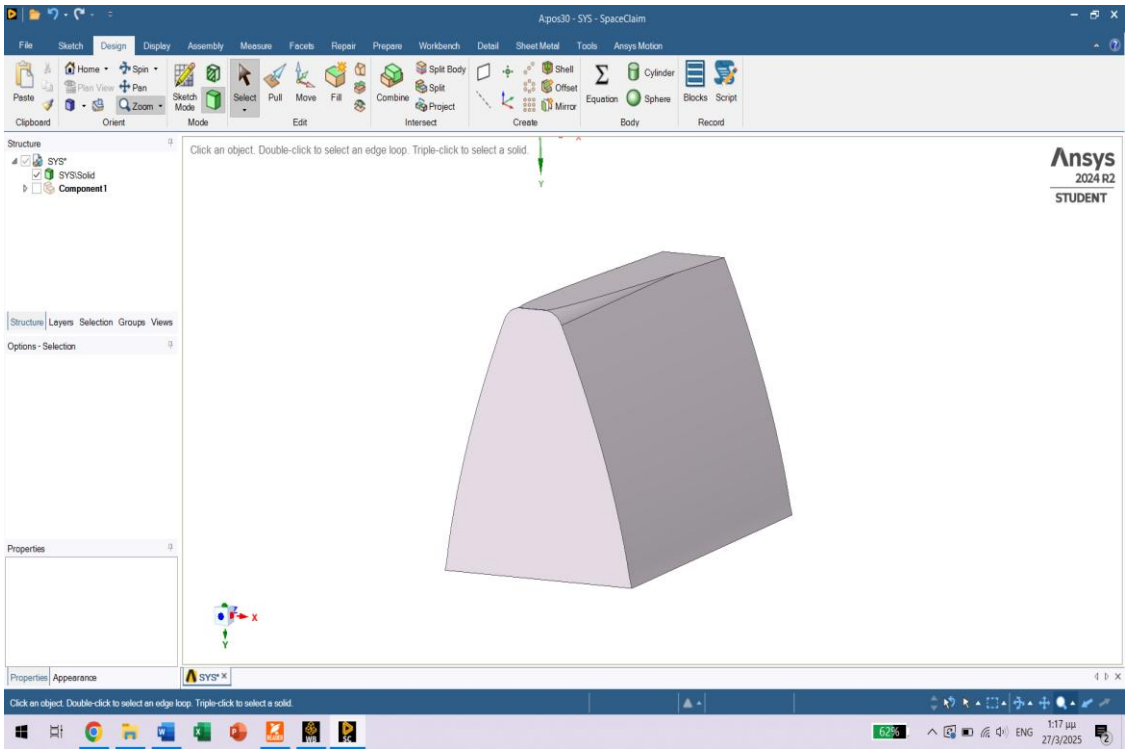


Figure 4.4 Final form of the cutting tooth

4.3 Chip geometry

After completing the cutting tool tooth, I follow the same procedure for the chips. To ensure an accurate and comprehensive analysis, as well as the production of reliable and well-documented results, I chose to use 9 different chips, each corresponding to a different cutting region. This choice allows the evaluation of the chips' effects at each point of the process, ensuring the highest possible accuracy and completeness in capturing the study's results.

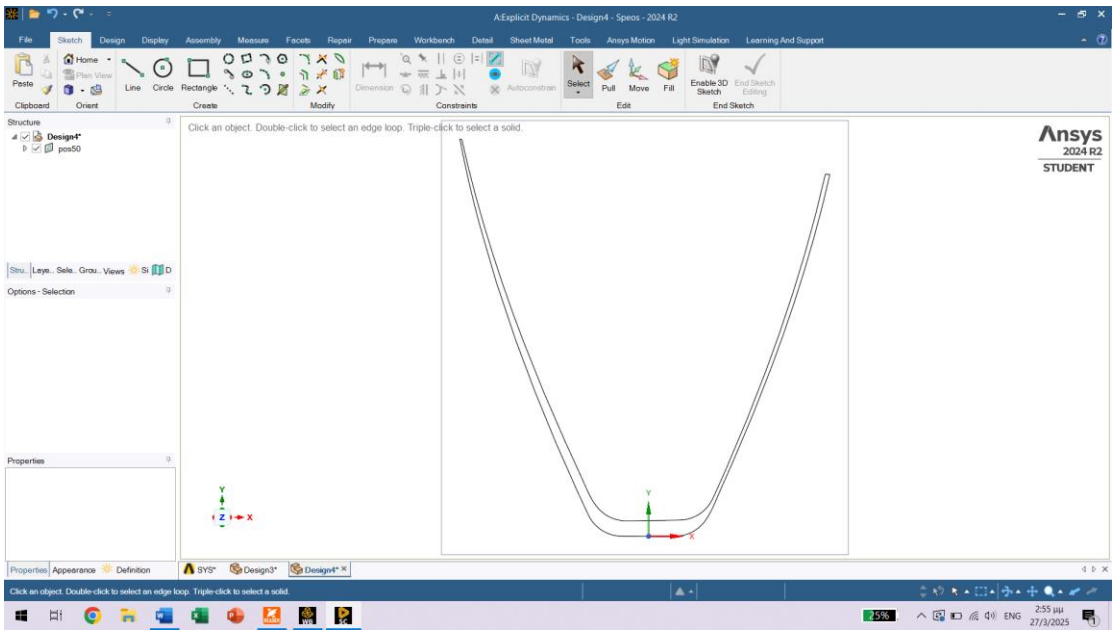


Figure 4.5 2D drawing of the chip

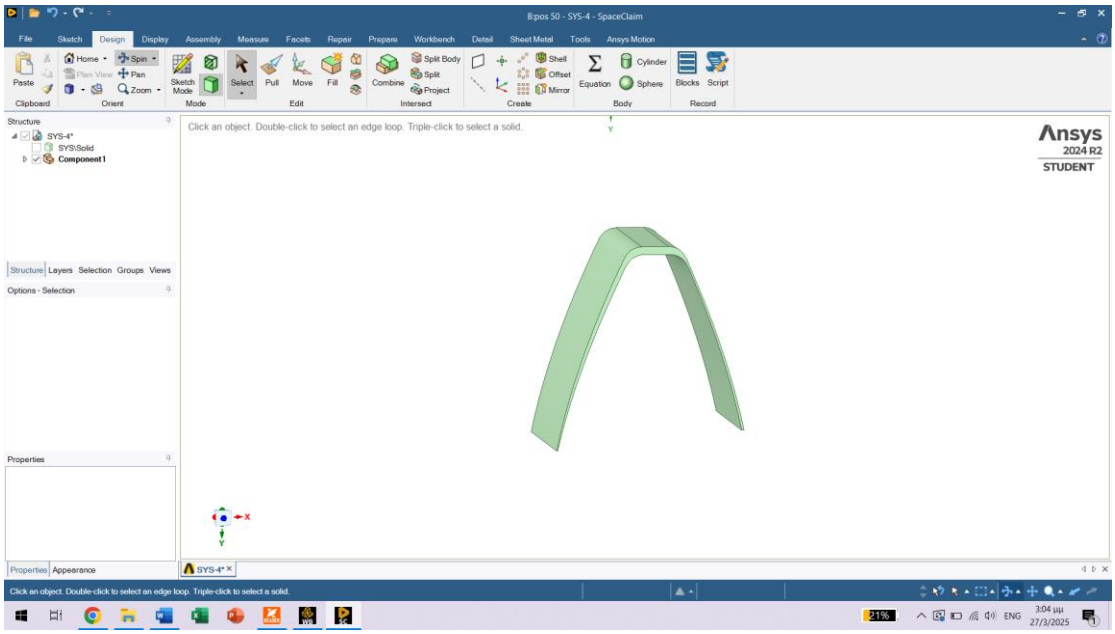


Figure 4.6 3D representation

After completing the three-dimensional design and fully shaping the geometric elements, the next and final step is the union of the two bodies, in order to proceed with the meshing process and the final analysis of the mechanical properties during machining. For this purpose, I use the Import function, through which I incorporate the chip design into the file containing the cutting tool model. Then, by selecting Combine, I precisely place the chip on the tool surface, thus completing the merging of the two elements and creating a single geometric unit, ready for the next phase of the analysis.

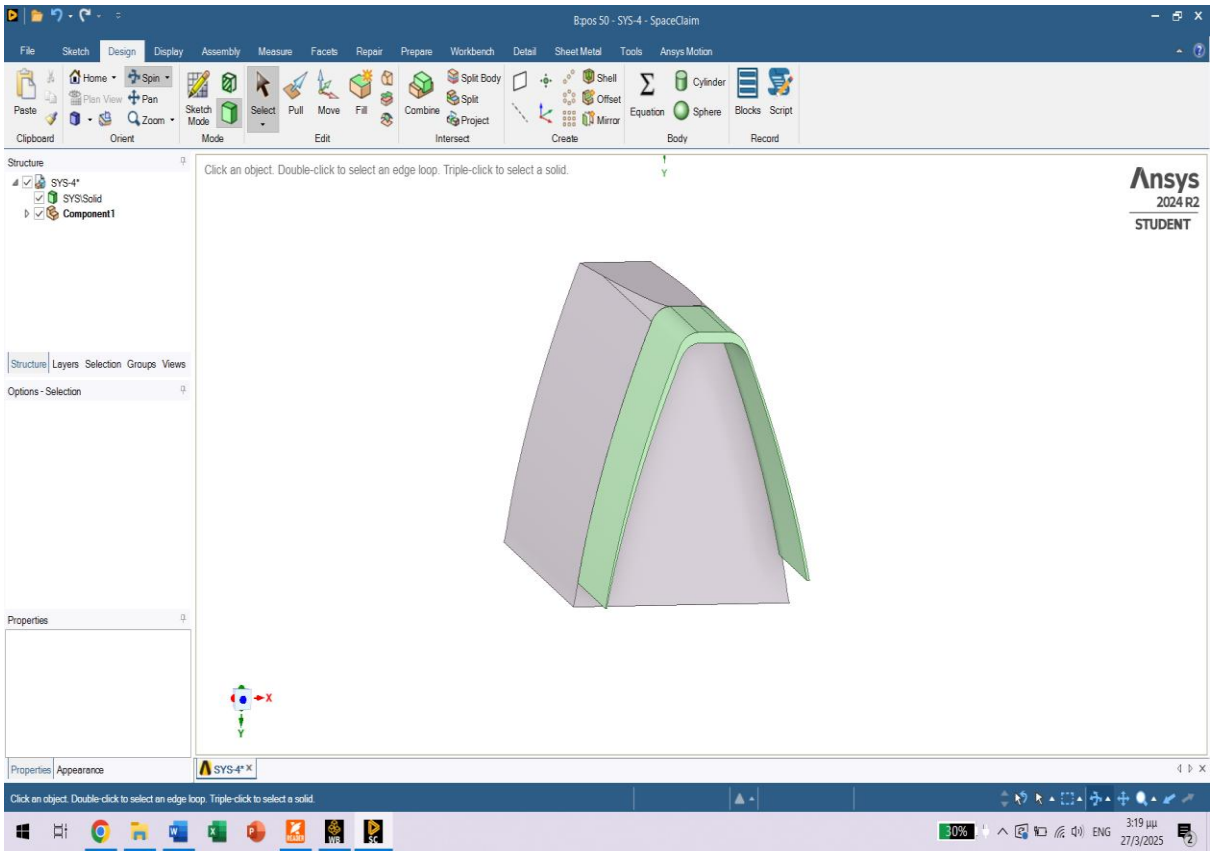


Figure 4.7 These two bodies merged into a single geometry

4.4 Meshing

To be able to analyze the stresses and deformations, I used the Static structural mode, which perfectly fits the needs and requirements of the thesis.

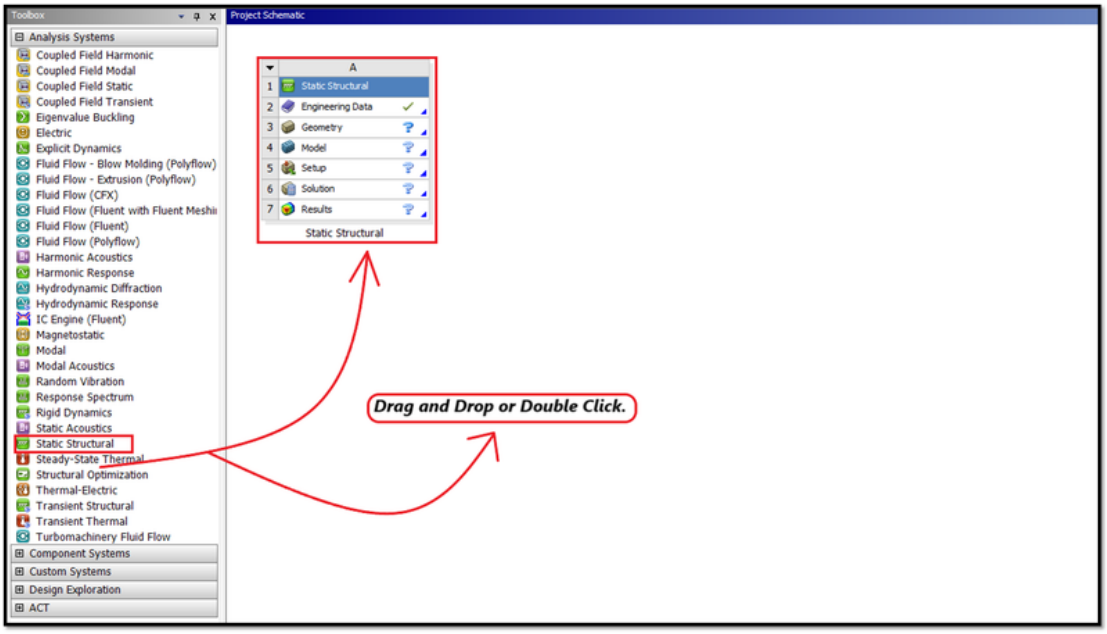


Figure 4.8 Selection of Static structural

Dividing the model into finite elements is one of the main stages in solving this type of problem. Regarding the cutting tool, which maintains the same geometry, the type and density of the meshing do not change; the point with the highest density is on the cutting edge. However, due to the different shape of the chips in each cutting position, the meshing varies according to the respective needs. In any case, the type of Mesh affects both the quality of the simulation and the solution time. Furthermore, the version I am using has a limit of 128,000 nodes, a fact I had to consider.

For example, in cutting position 40, the part of the chip located at the bottom right is very thin; consequently, the meshing elements in this specific area need to be considerably smaller than in the rest of the body. In this specific case, the size of the smallest element is 0.029816 mm.

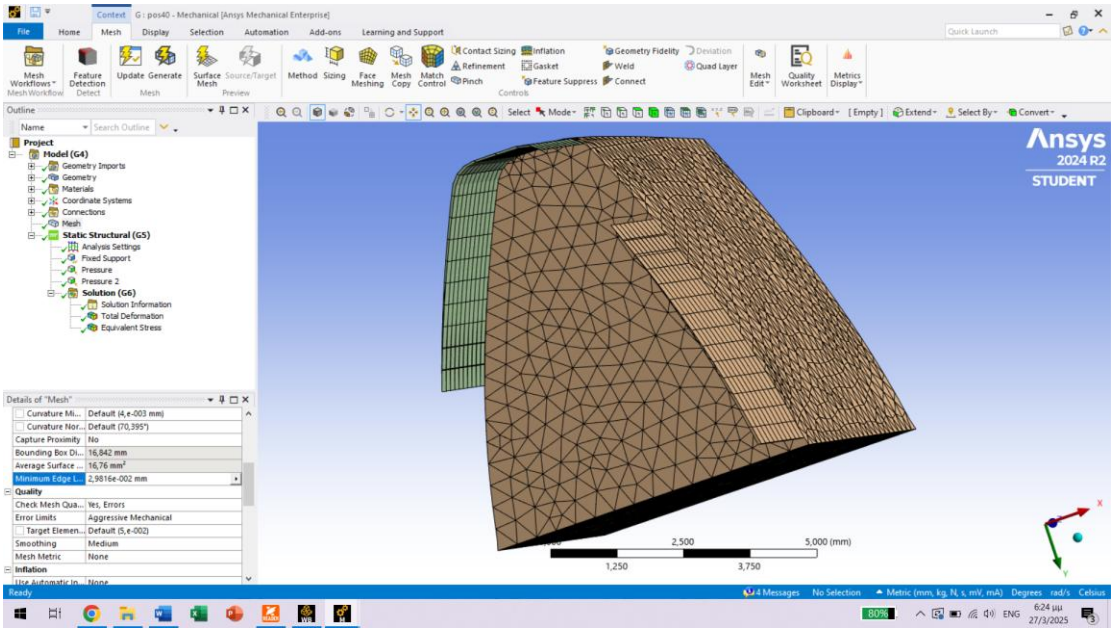


Figure 4.9 Meshing at position 40

4.5 Material definition

The materials selected are widely used in the industry. For the cutting tool, High-speed steel was chosen, and for the chip, 16MnCr5 (16 Manganese, 5 Chromium). These specific materials were not present in the program's material library, so I had to create them using some of their basic parameters such as Young's modulus (E), density (ρ), and Poisson's ratio (v).

	hss	16MNcR5
Poisson (v)	0,28	0,29
E (GPa)	233	210
p (kg/m³)	7870	7850

Table 4.1 Material parameters

To be able to introduce the new materials into the software, I must open Engineering Data and, in the "add new material" field, define the parameters mentioned above.

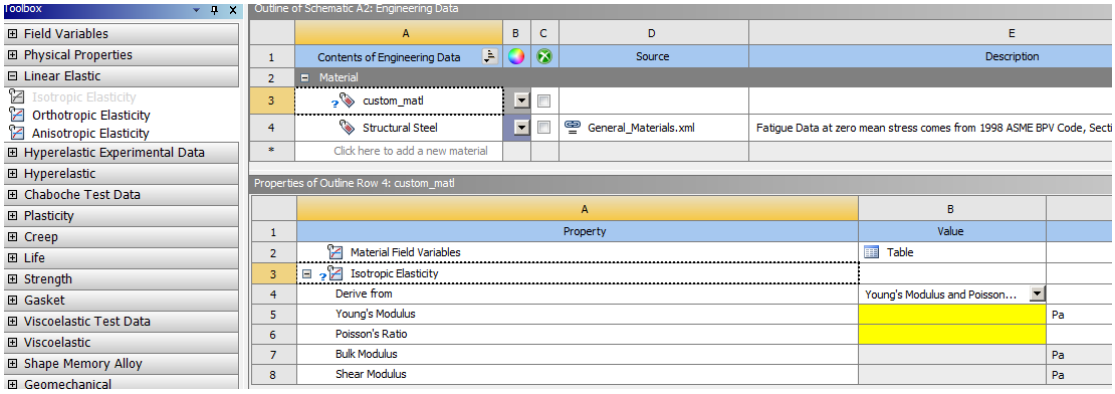


Figure 4.10 Engineering data menu

4.6 Fixing and Pressure

In analysis problems using the finite element method, the correct application of boundary conditions is necessary for the realistic simulation of a structure's mechanical behavior.

In this specific thesis, the base of the cutting tool has been fixed, meaning the nodes at this point cannot translate or rotate in the three axes. This choice is critical, as it ensures that the tip of the tool remains stationary, as would also happen in a real fixing condition. Furthermore, the fixing helps avoid numerical errors in the simulation, ensuring that the analysis runs correctly and yields reliable results.

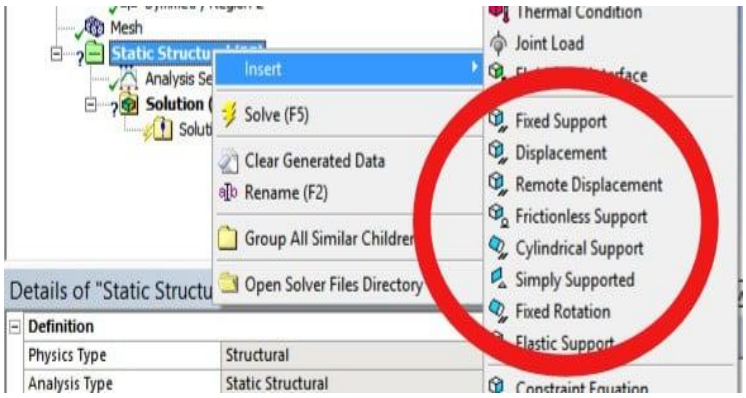


Figure 4.11 Application of fixed support

To apply the fixing, I select Static structural and then insert and fixed support, as shown in Figure 4.11.

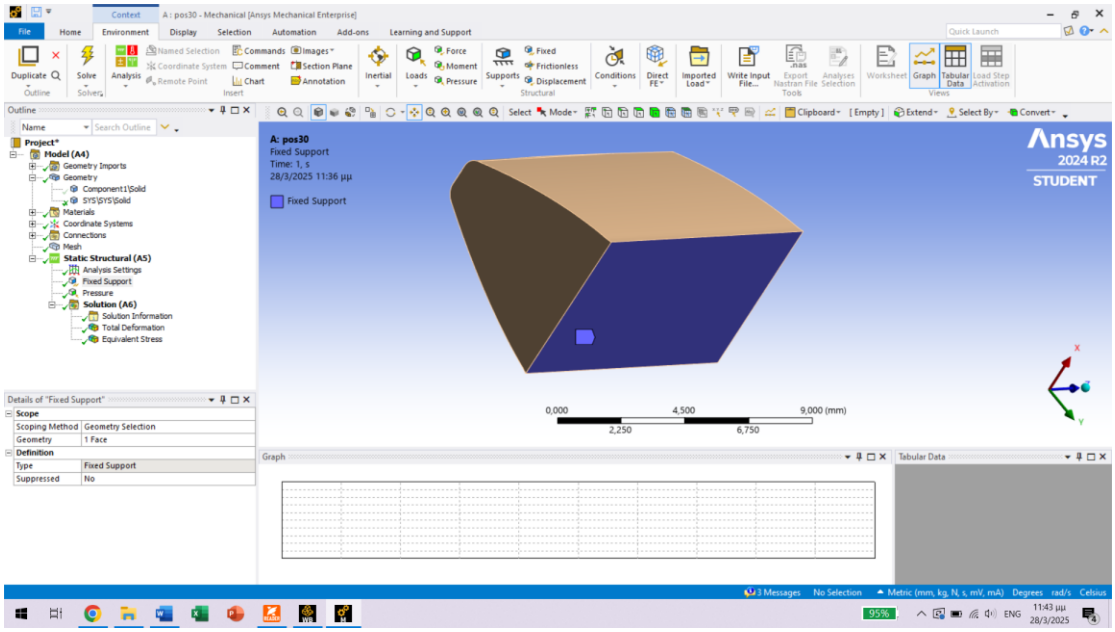


Figure 4.12 Fixing at the base of the cutting tool

The last factor to be defined before starting the simulation is pressure, which has been set at 1500 N/mm^2. The pressure will be applied to the chip, perpendicular to the cutting tool surface, in the area near the cutting edge. To introduce it, I select Static Structural and Insert Pressure.

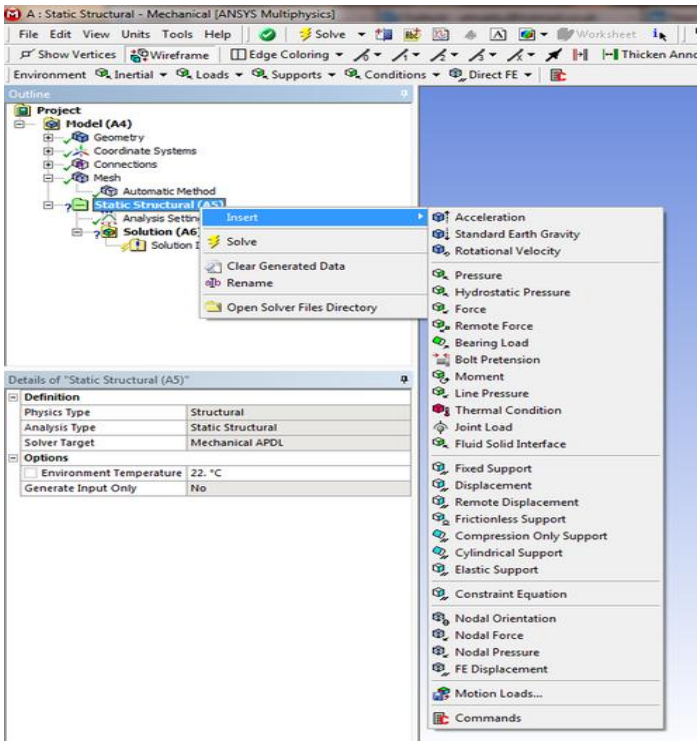


Figure 4.13 Application of pressure

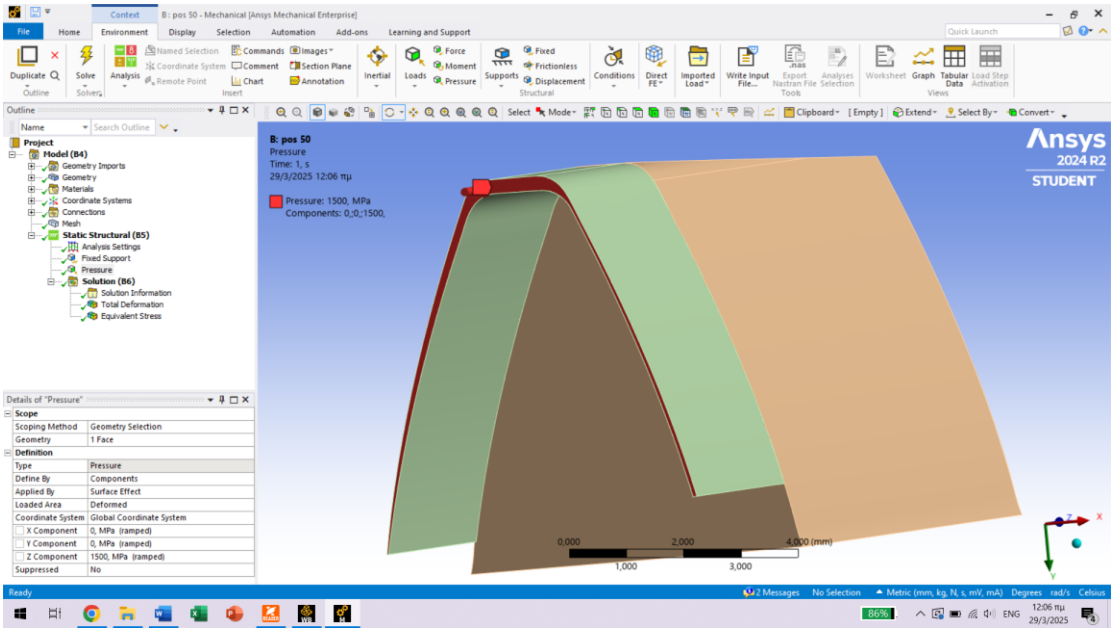


Figure 4.14 Application of pressure to the chip

4.7 Stress and strain calculation

After defining all parameters, I can start the simulation by going to Model → Static structural → Solve. The time required to complete the simulation depends on the meshing and the capabilities of the processor on which the program is running.

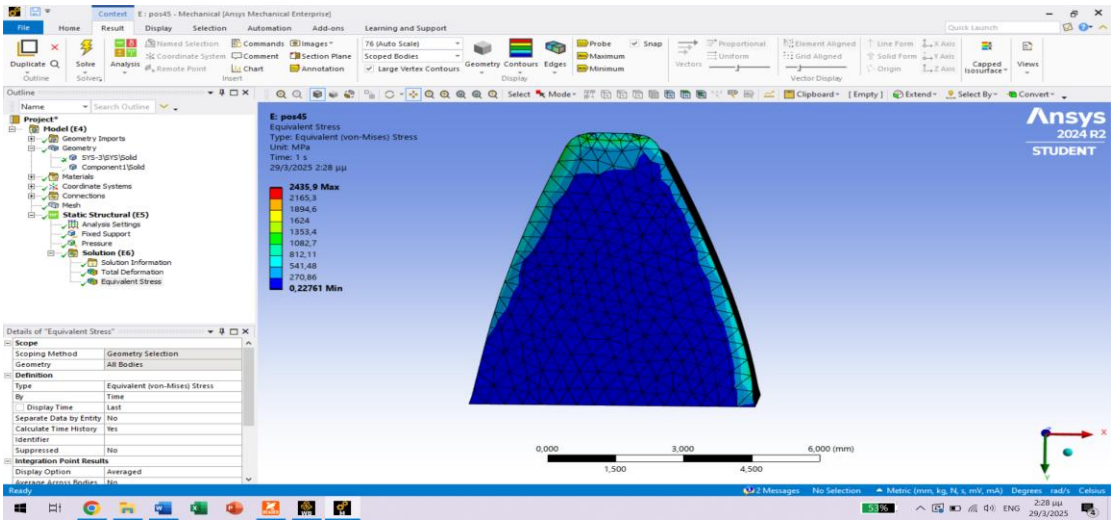


Figure 4.15 Simulation result

In the processing area, the model is displayed, with stresses rendered in color: low values appear in blue, while high values appear in red. The option to display additional results, such as nodal displacements and principal stresses, is also provided. Furthermore, the user can create graphs by selecting specific nodes for analysis.

5. RESULTS

This chapter presents the results obtained from the analysis using the finite element program. For this purpose, nine files were created, each corresponding to a different cutting position, from position 30 to position 70.

For each cutting position, a diagram was created depicting the Von Mises stresses along the cutting edge. Additionally, a summary diagram with the maximum Von Mises stresses for each cutting position is provided, as well as diagrams presenting the deformation in each of the examined positions.

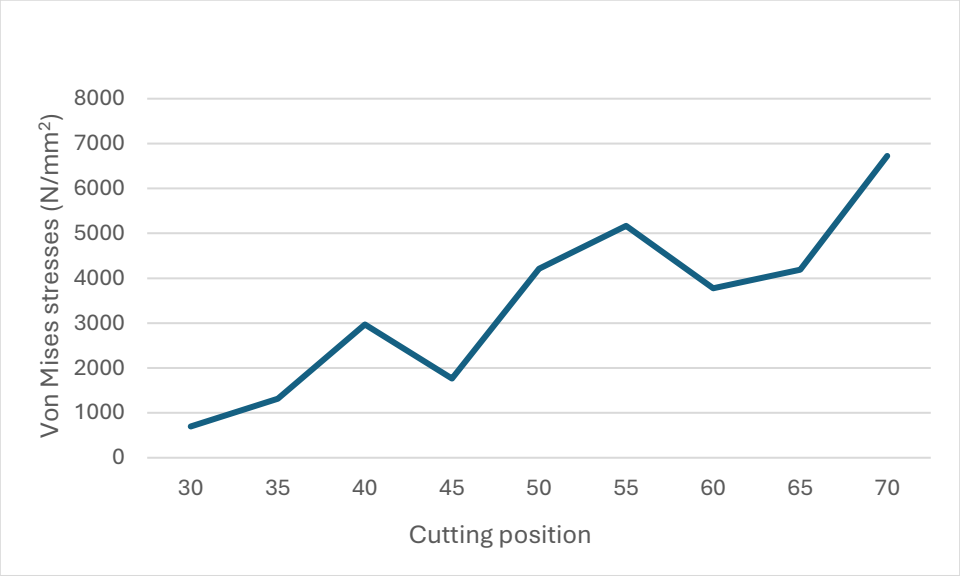


Figure 5.1 Maximum Von Mises stresses for each cutting position

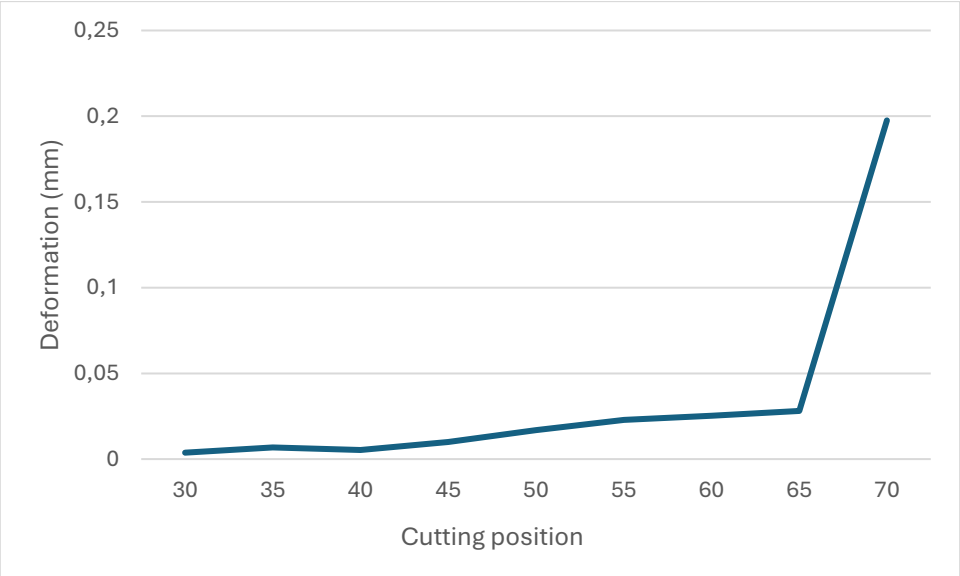


Figure 5.2 Maximum deformation for each cutting position

Position 30

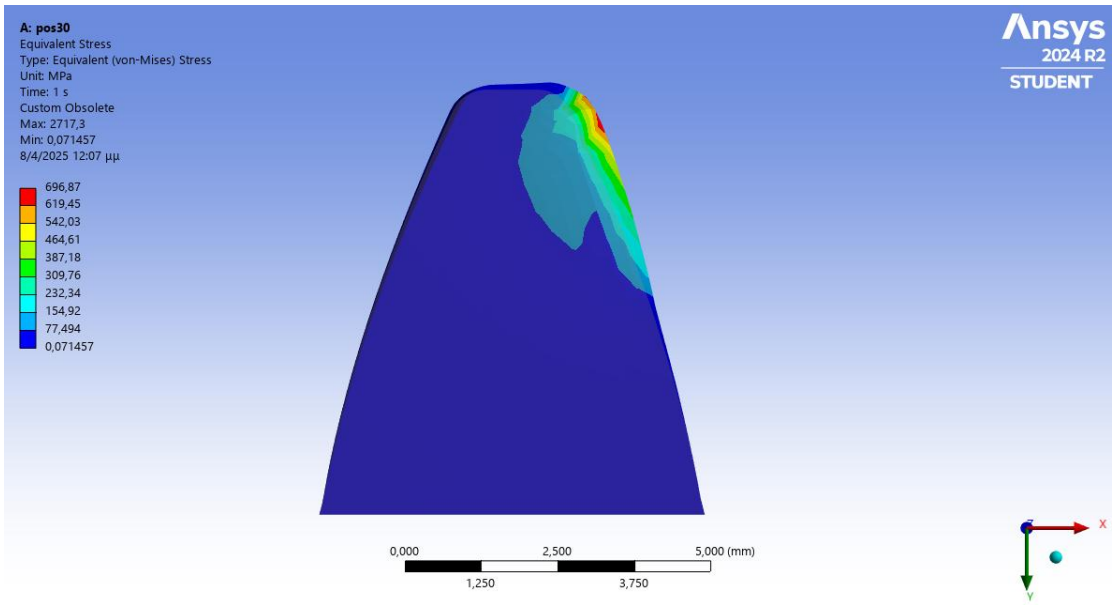


Figure 5.3 Von Mises stresses for cutting position 30

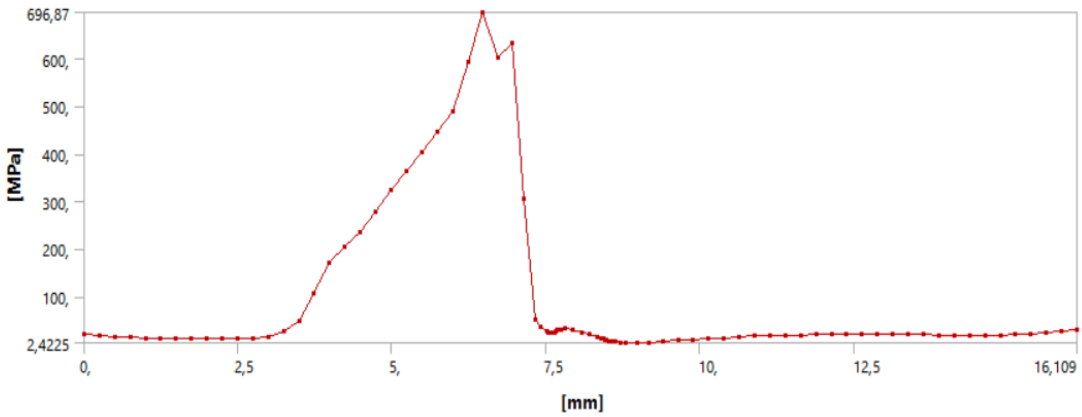


Figure 5.4 Von Mises stresses along the cutting edge, for position 30

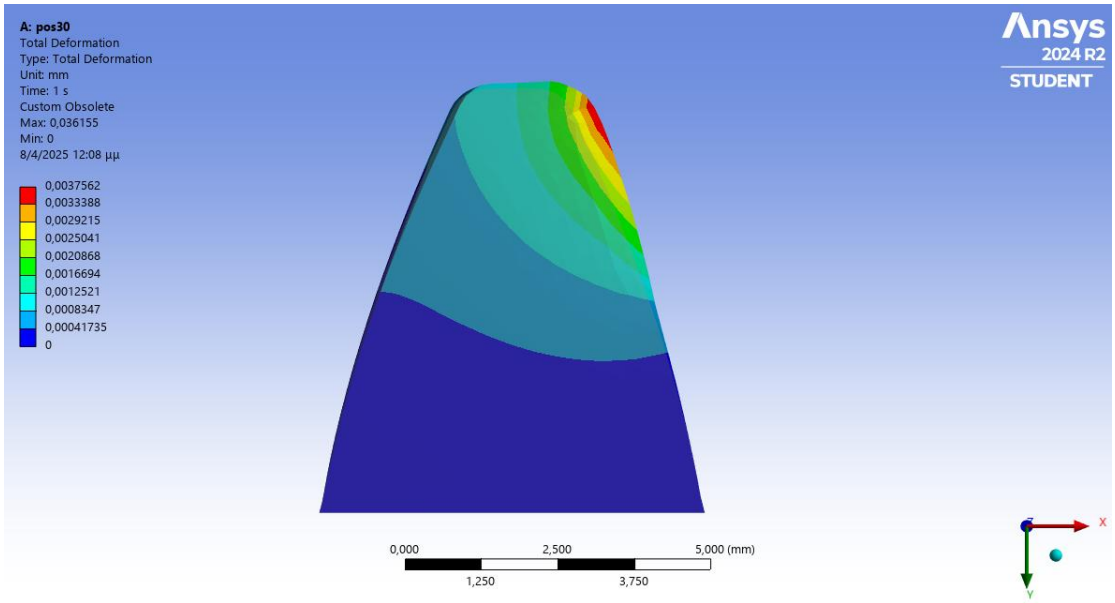


Figure 5.5 Deformation of the tool for cutting position 30

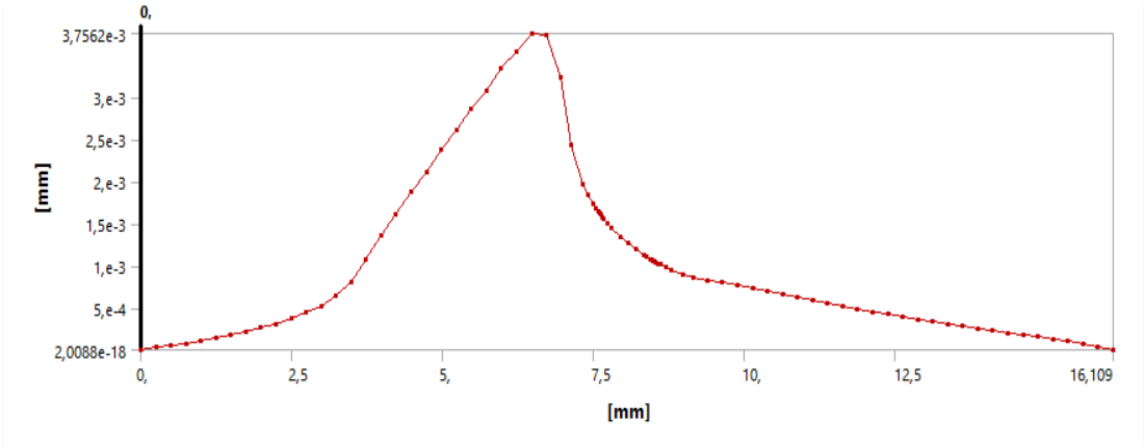


Figure 5.6 Deformation along the cutting edge, for position 30

Position 35

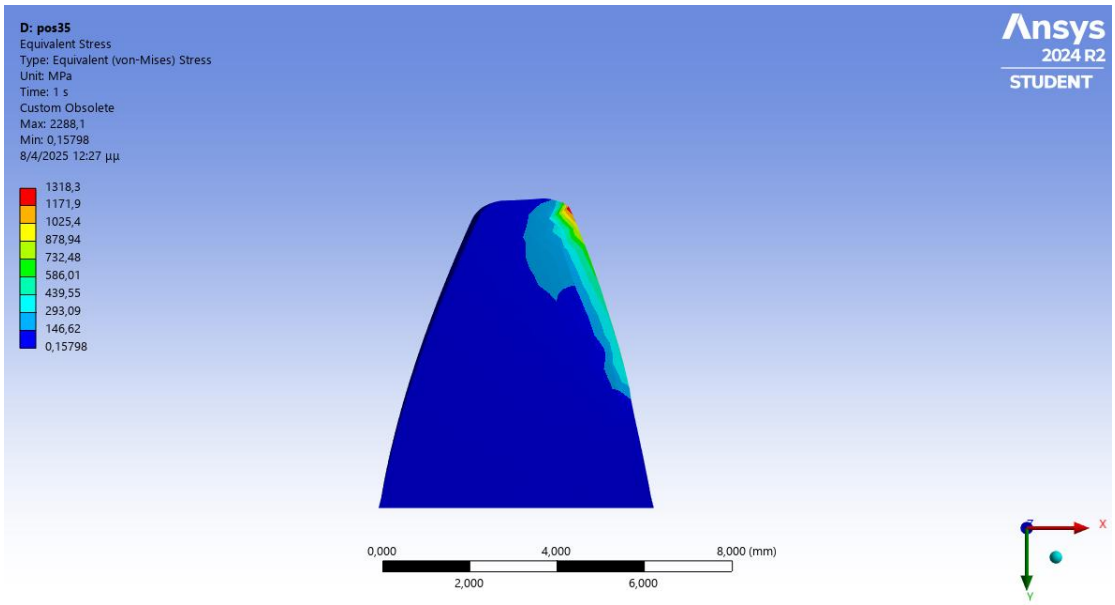


Figure 5.7 Von Mises stresses for cutting position 35

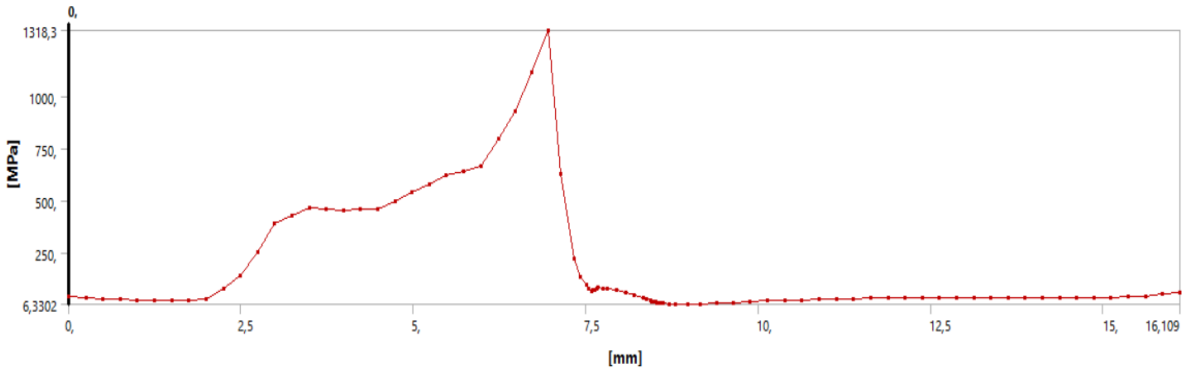


Figure 5.8 Von Mises stresses along the cutting edge, for position 35

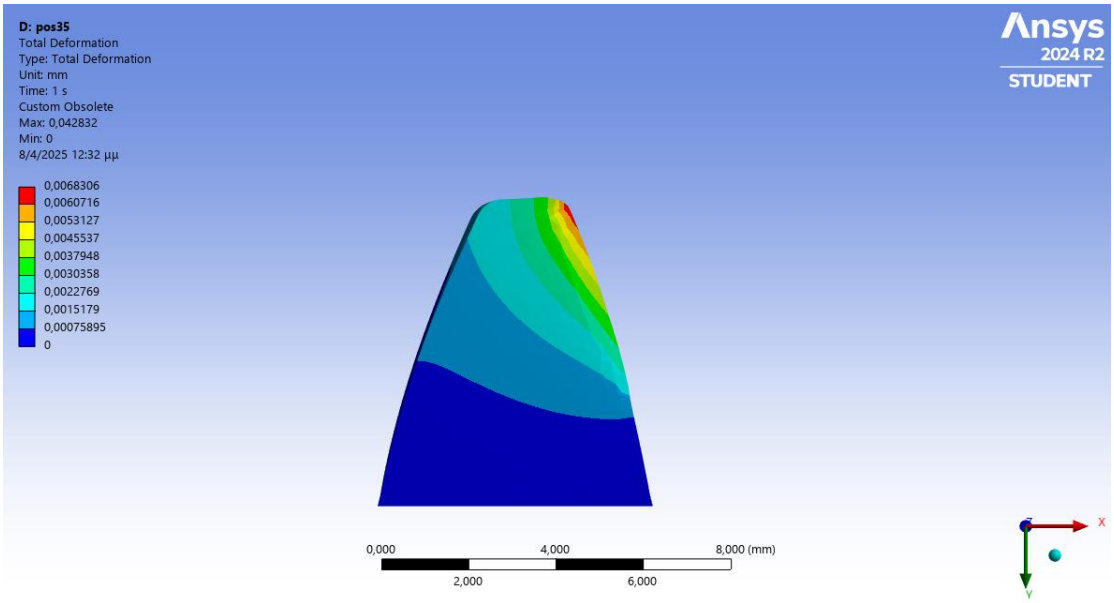


Figure 5.9 Deformation of the tool for cutting position 35

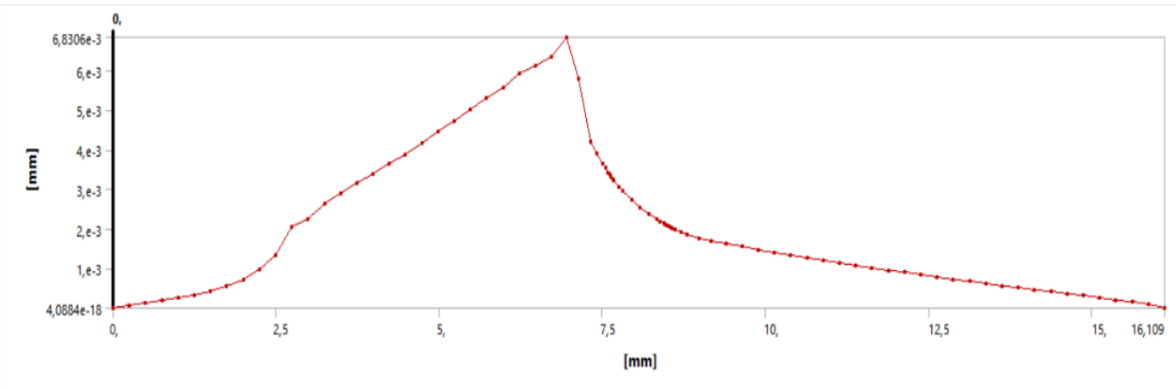


Figure 5.10 Deformation along the cutting edge, for position 35

Position 40

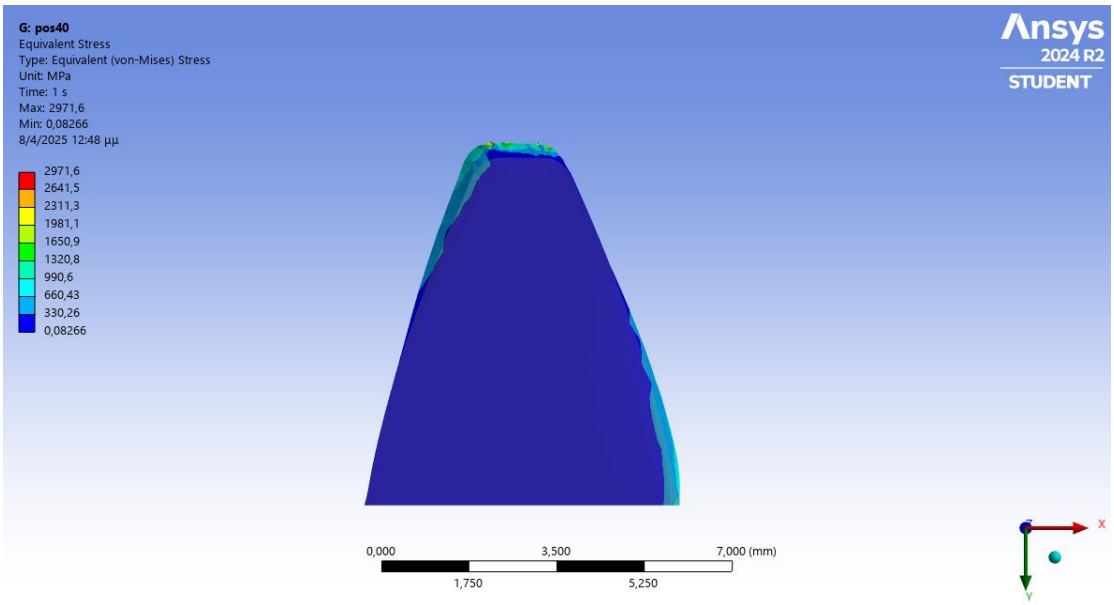


Figure 5.11 Von Mises stresses for cutting position 40

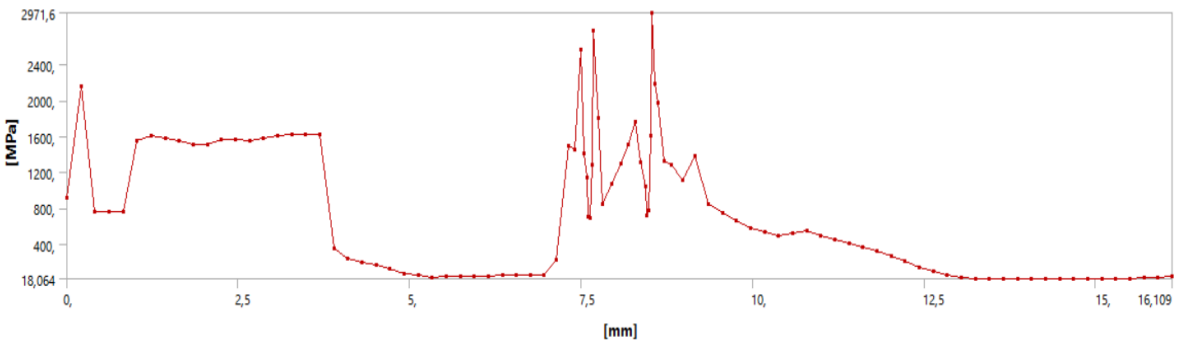


Figure 5.12 Von Mises stresses along the cutting edge, for position 40

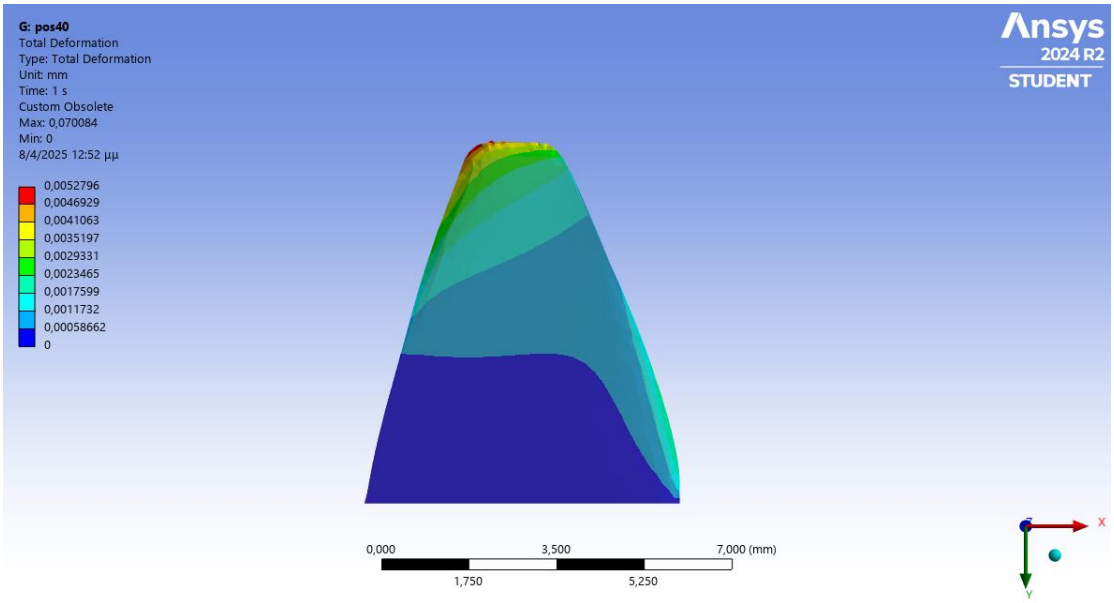


Figure 5.13 Deformation of the tool for cutting position 40

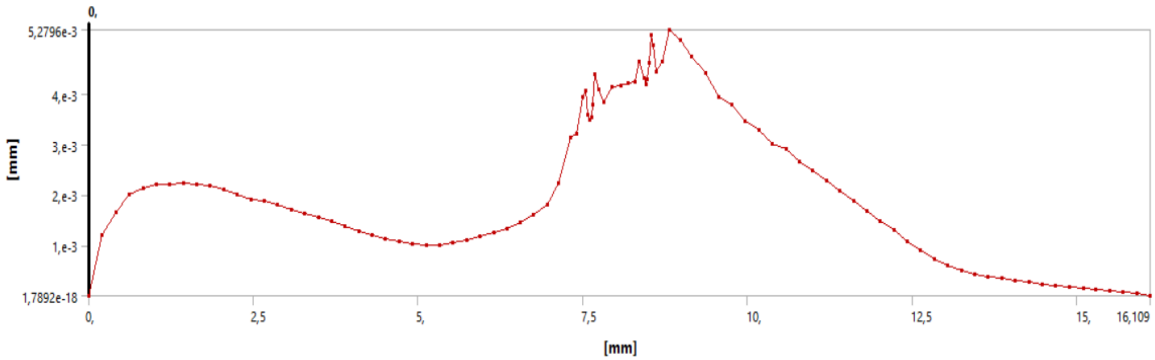


Figure 5.14 Deformation along the cutting edge, for position 40

Position 45

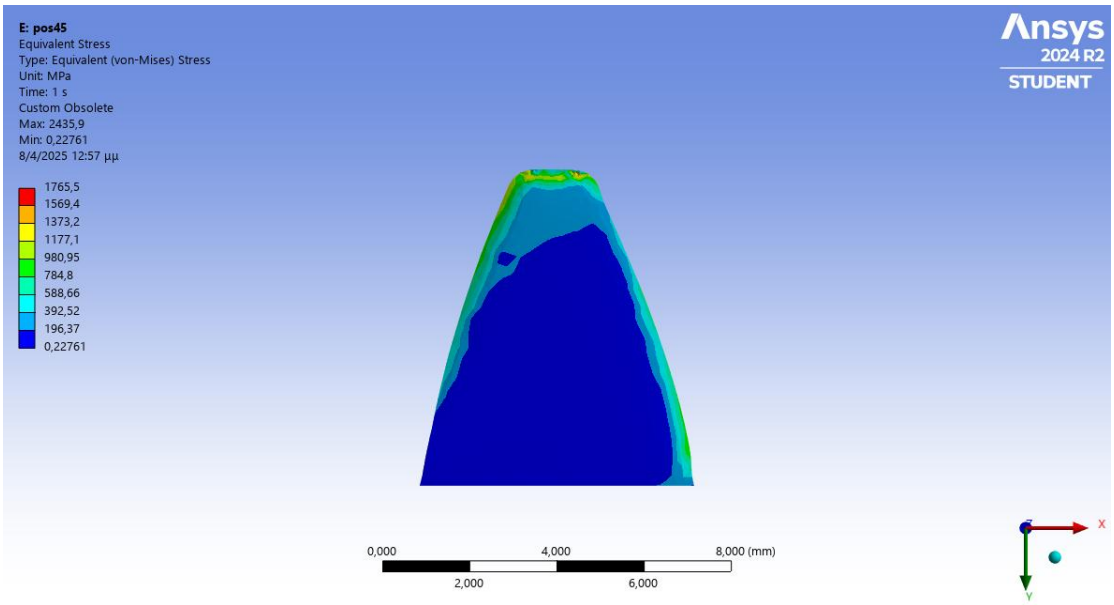


Figure 5.15 Von Mises stresses for cutting position 45

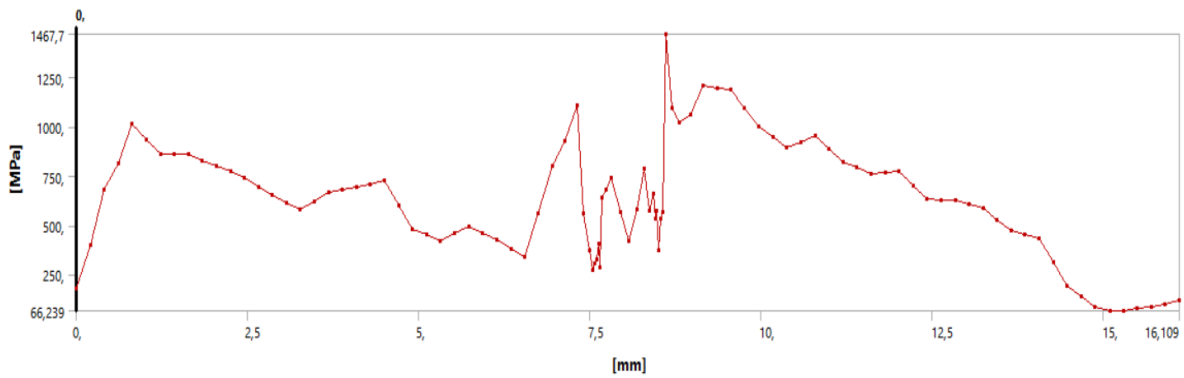


Figure 5.16 Von Mises stresses along the cutting edge, for position 45

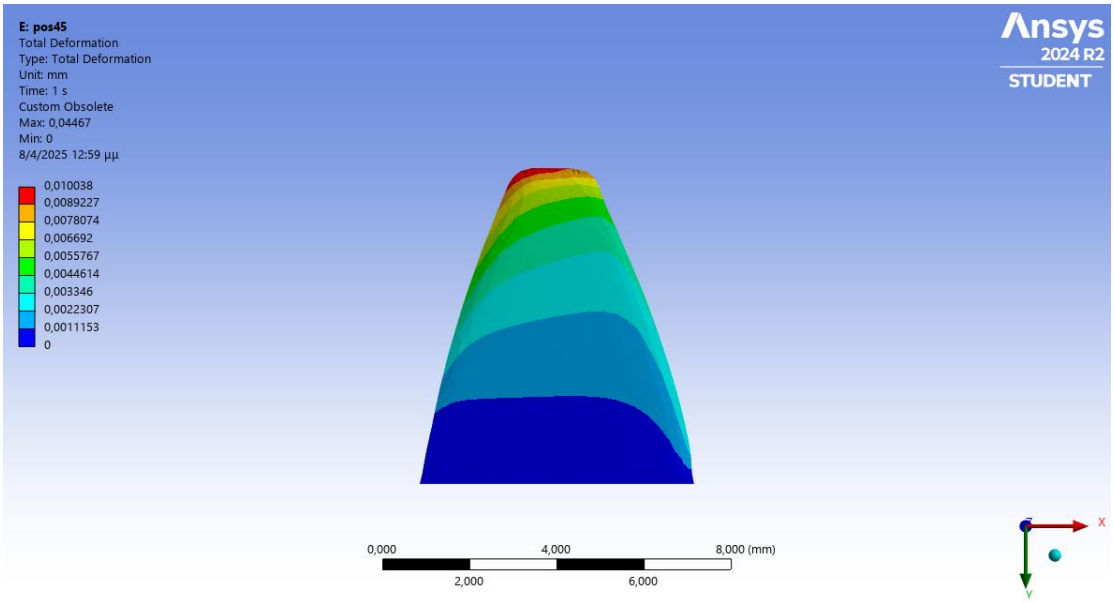


Figure 5.17 Deformation of the tool for cutting position 45

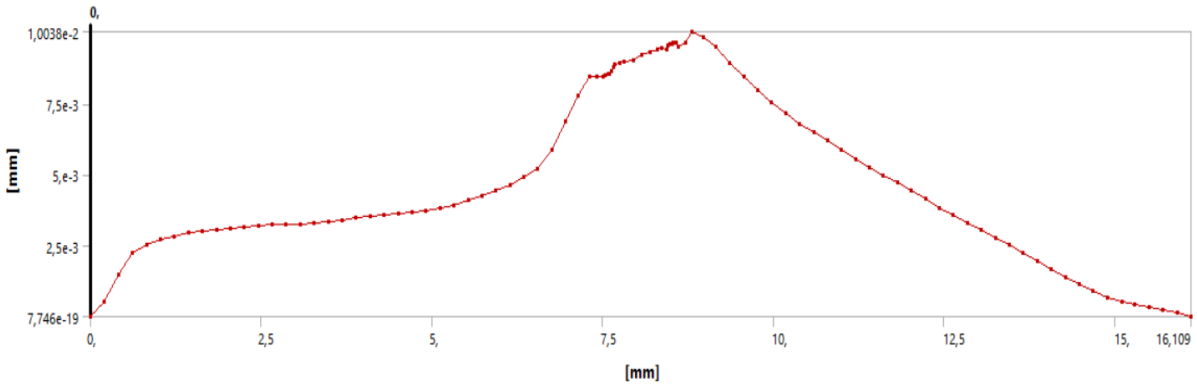


Figure 5.18 Deformation along the cutting edge, for position 45

Position 50

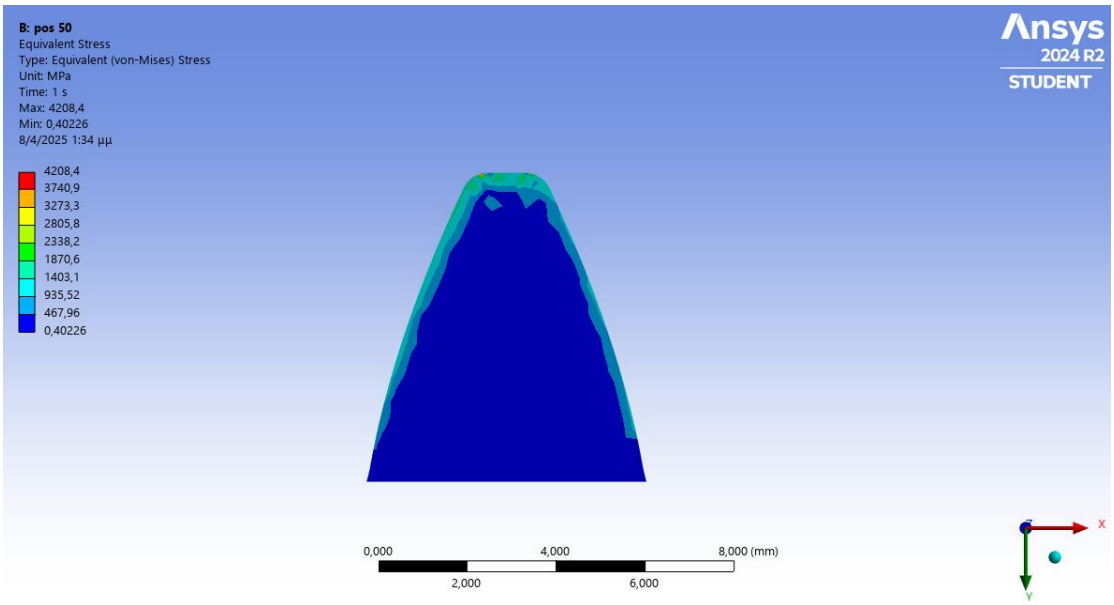


Figure 5.19 Von Mises stresses for cutting position 50

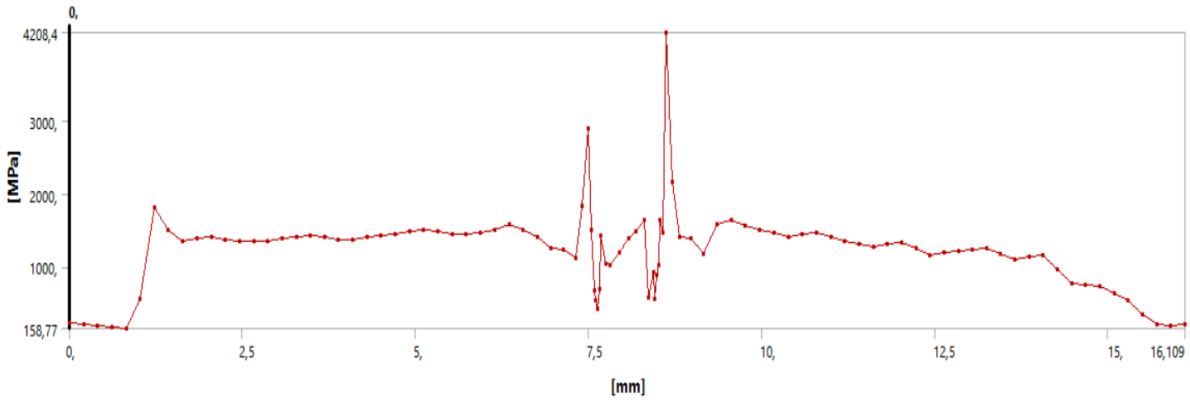


Figure 5.20 Von Mises stresses along the cutting edge, for position 50

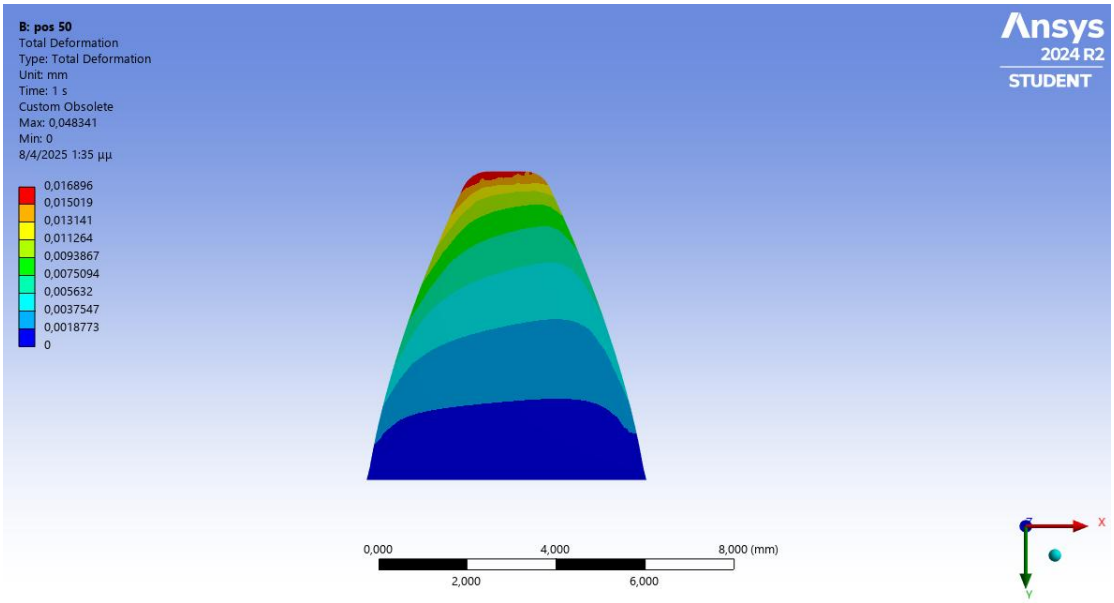


Figure 5.21 Deformation of the tool for cutting position 50

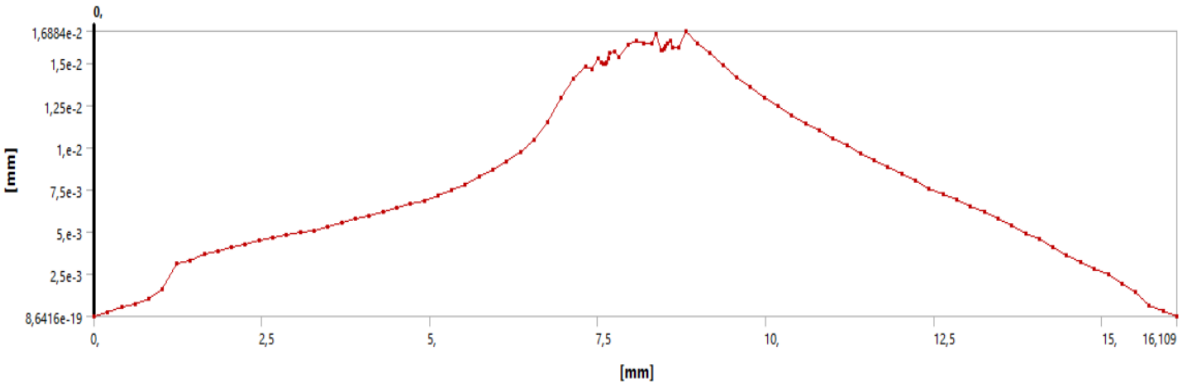


Figure 5.22 Deformation along the cutting edge, for position 50

Position 55

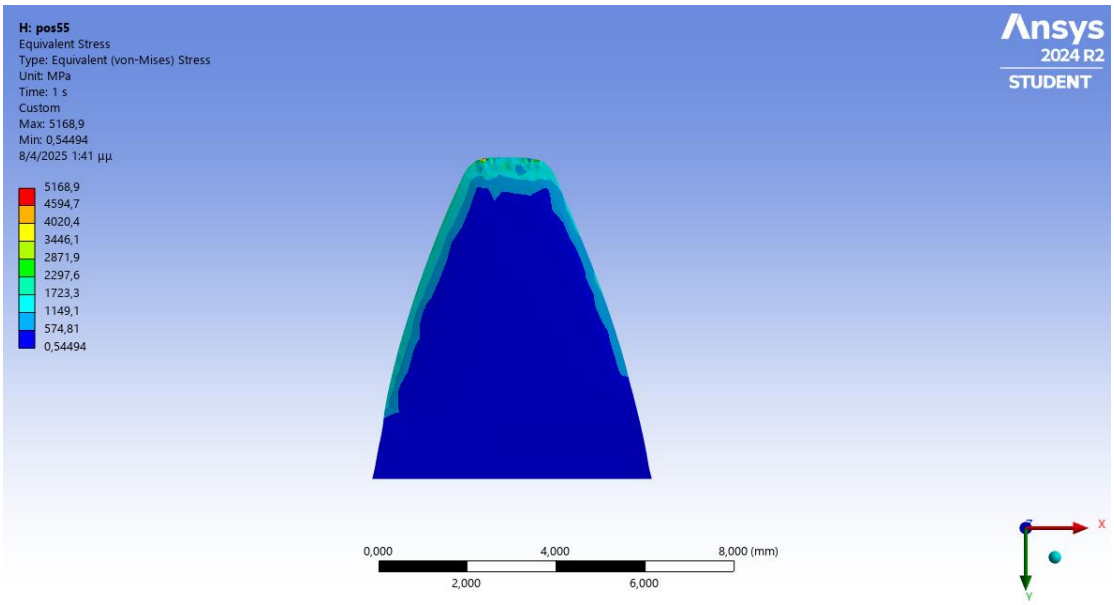


Figure 5.23 Von Mises stresses for cutting position 55

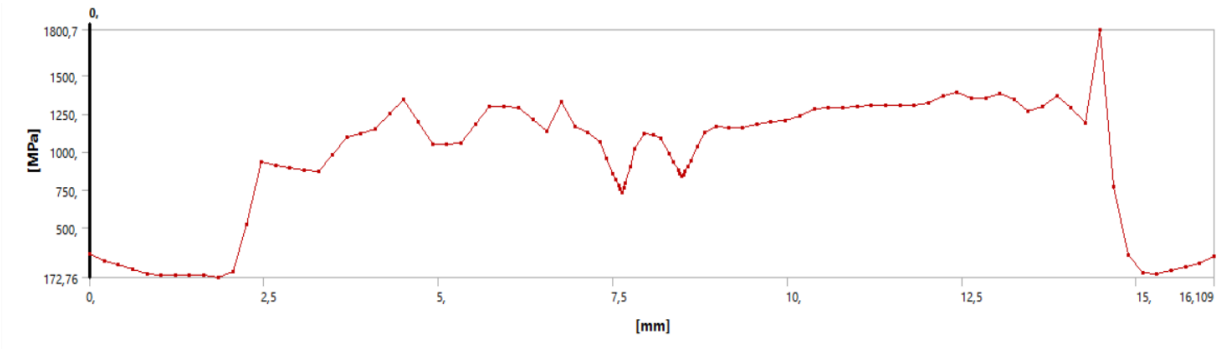


Figure 5.24 Von Mises stresses along the cutting edge, for position 55

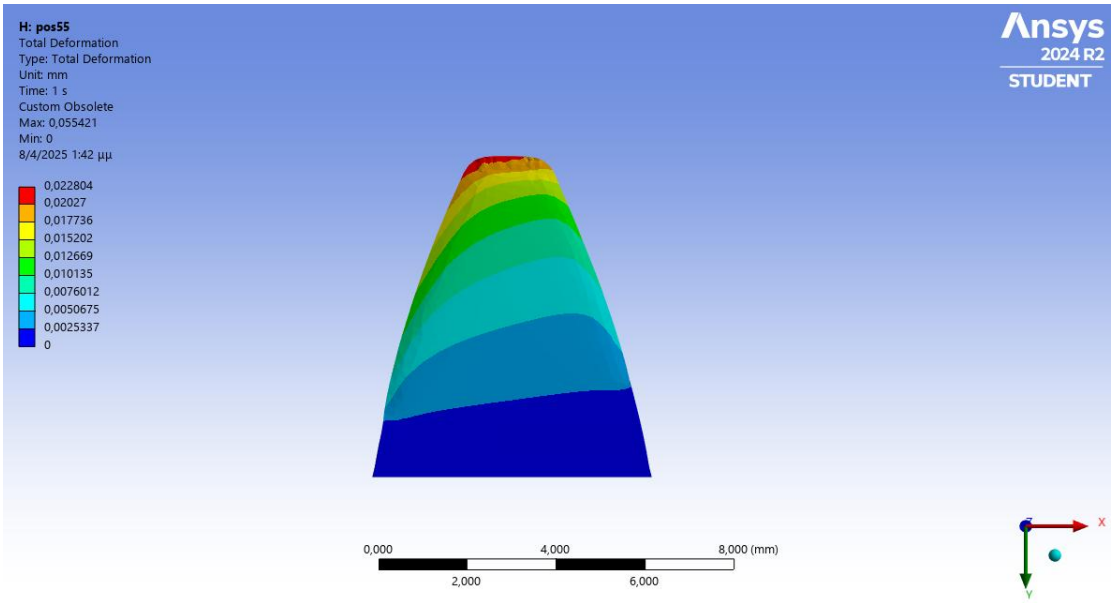


Figure 5.25 Deformation of the tool for cutting position 55

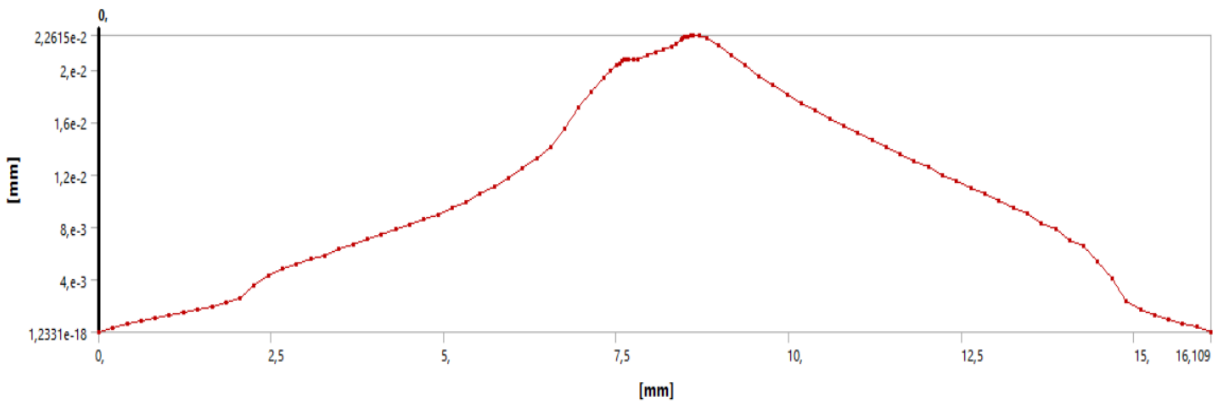


Figure 5.26 Deformation along the cutting edge, for position 55

Position 60

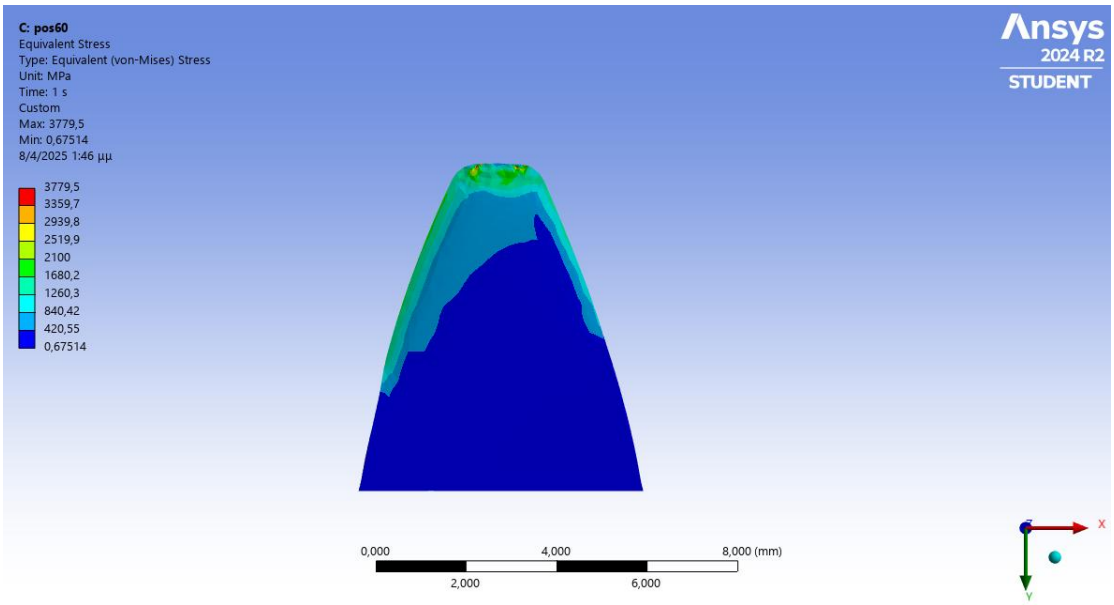


Figure 5.27 Von Mises stresses for cutting position 60

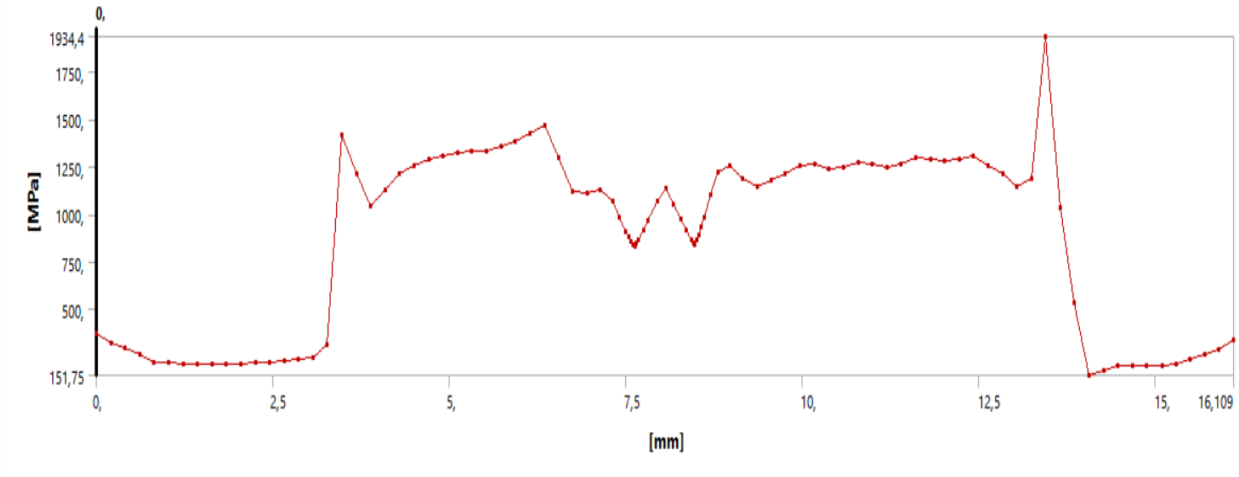


Figure 5.28 Von Mises stresses along the cutting edge, for position 60

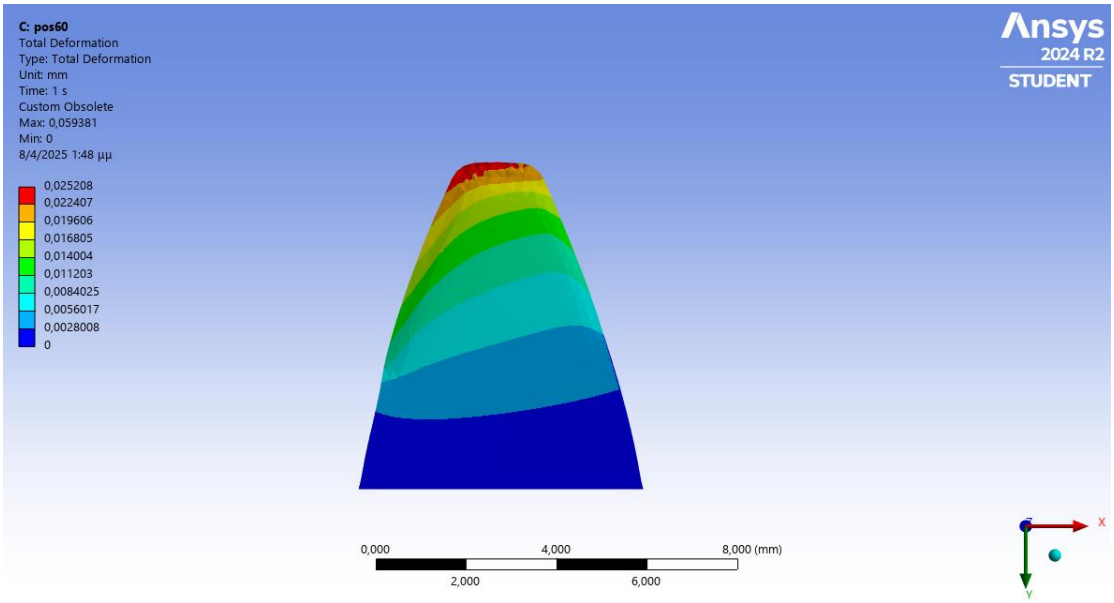


Figure 5.29 Deformation of the tool for cutting position 60

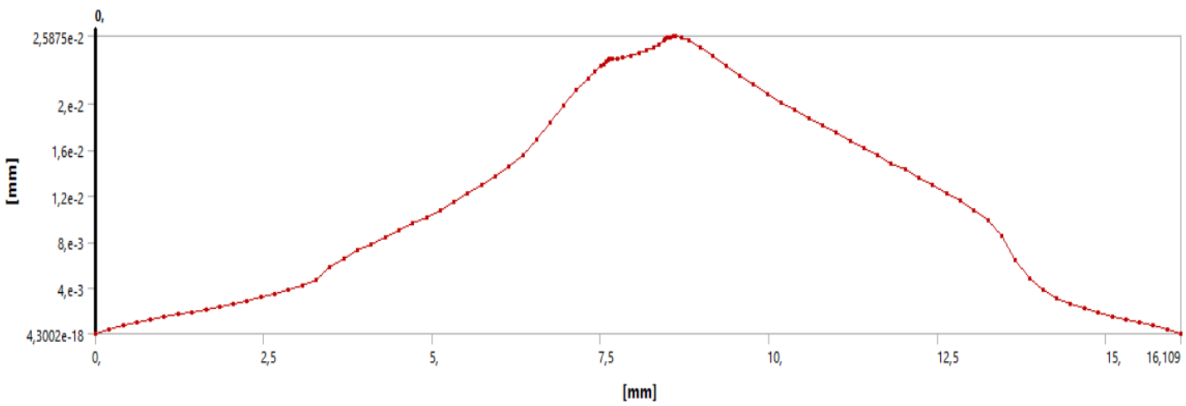


Figure 5.30 Deformation along the cutting edge, for position 60

Position 65

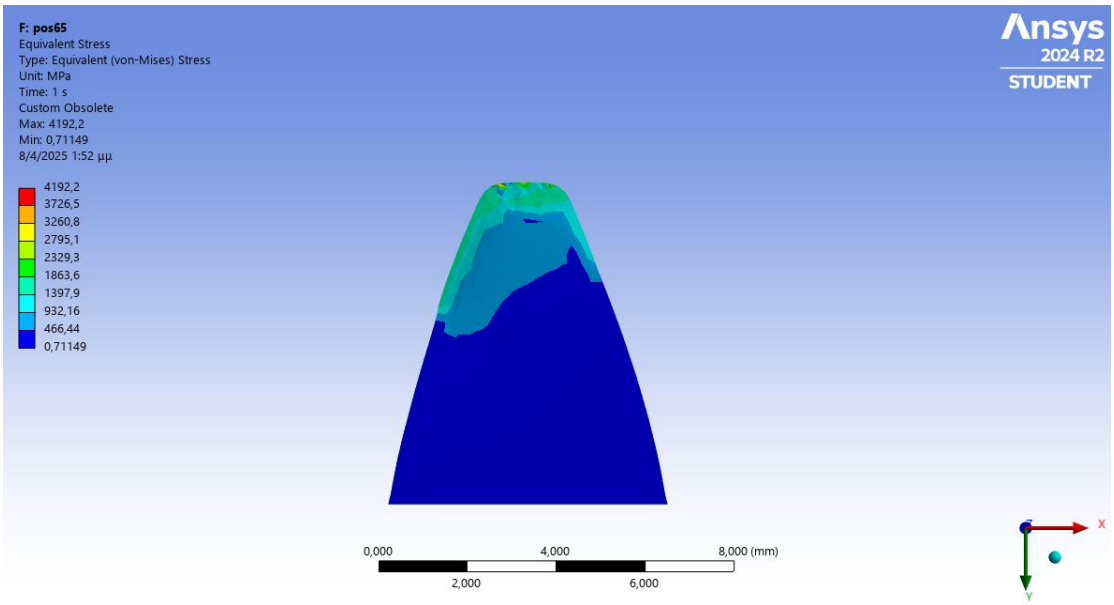


Figure 5.31 Von Mises stresses for cutting position 65

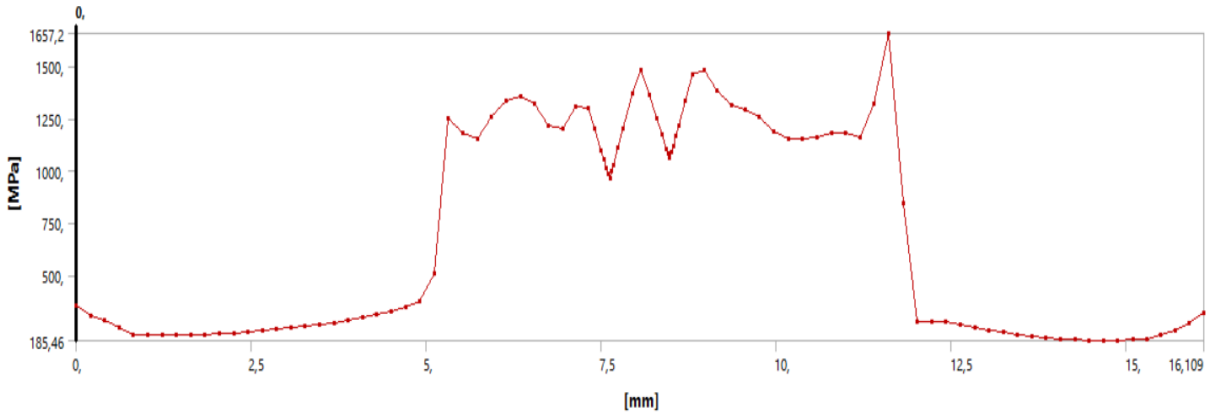


Figure 5.32 Von Mises stresses along the cutting edge, for position 65

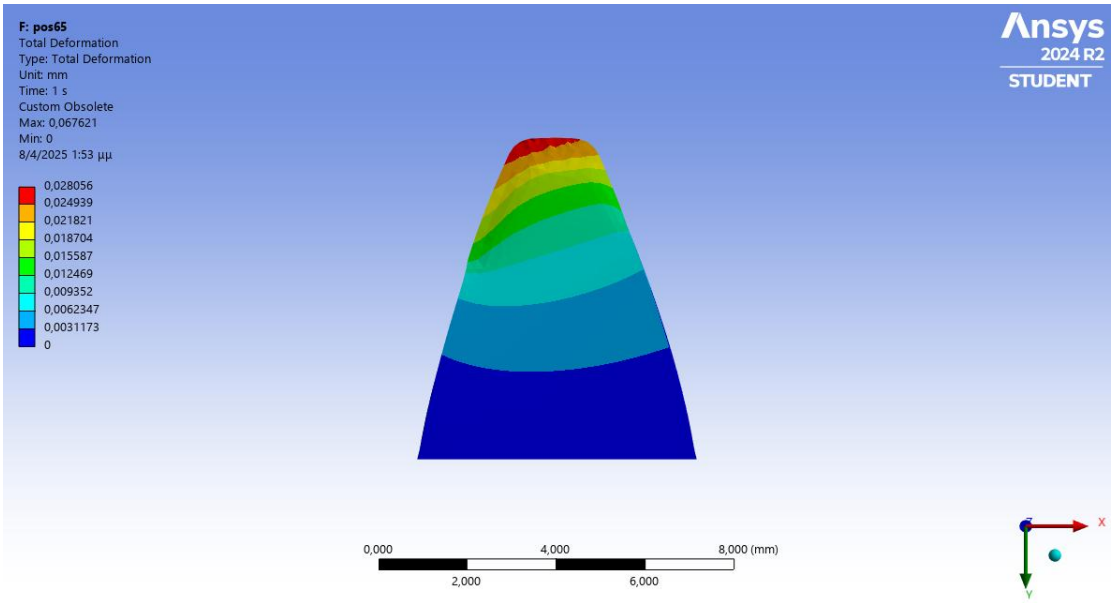


Figure 5.33 Deformation of the tool for cutting position 65

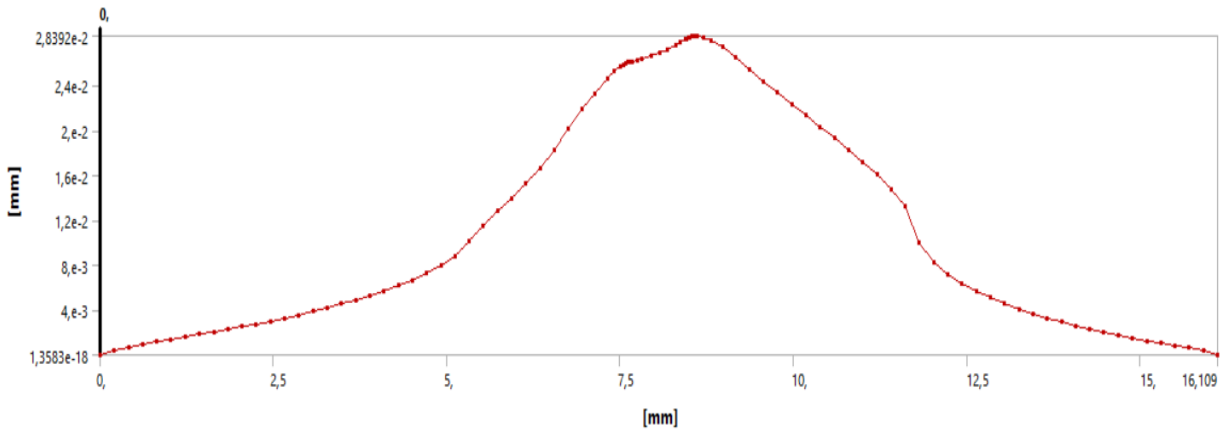


Figure 5.34 Deformation along the cutting edge, for position 65

Position 70

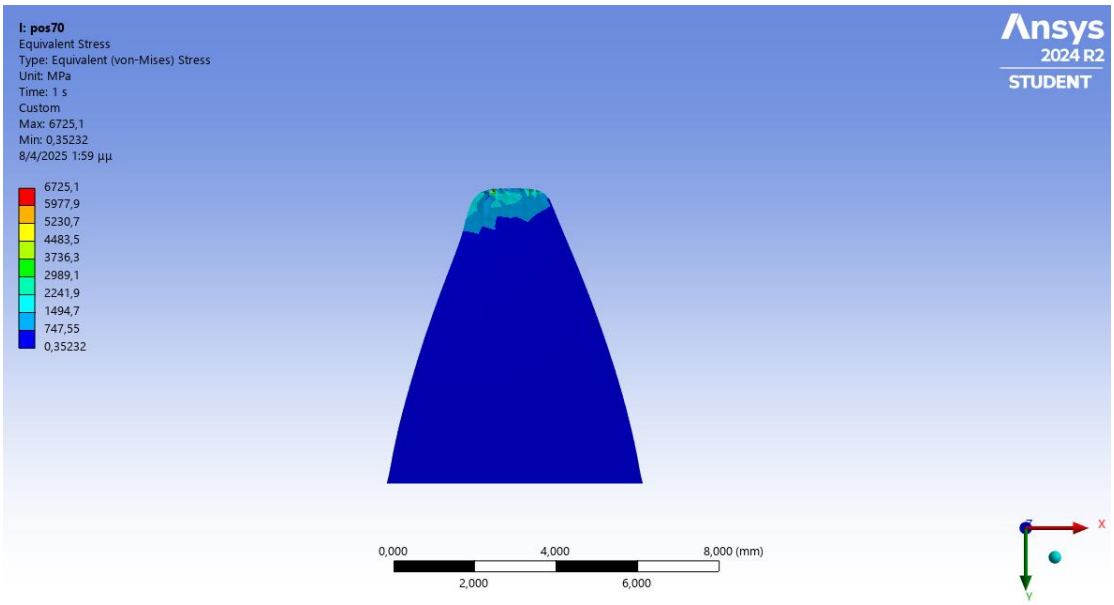


Figure 5.35 Von Mises stresses for cutting position 70

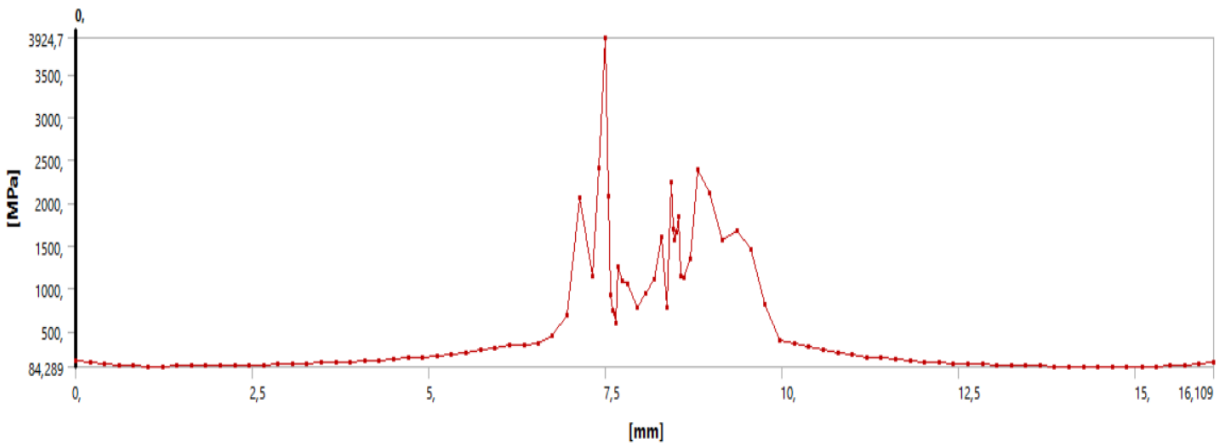


Figure 5.36 Von Mises stresses along the cutting edge, for position 70

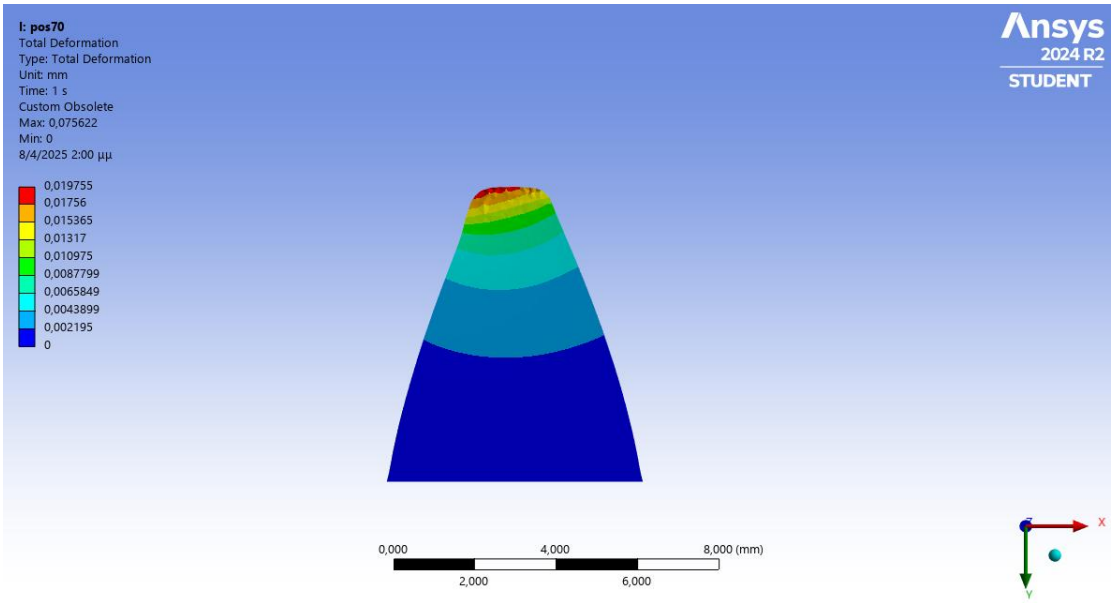


Figure 5.37 Deformation of the tool for cutting position 70

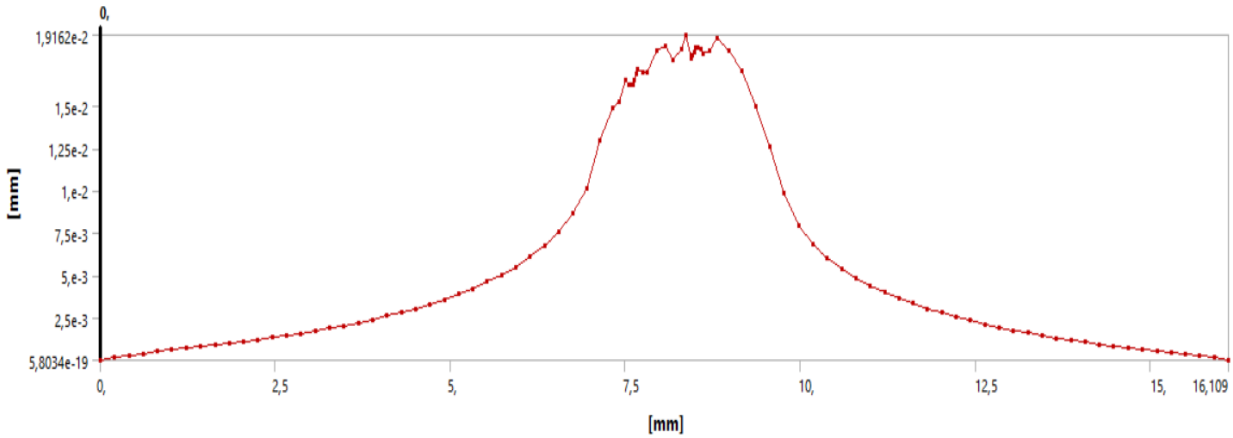


Figure 5.38 Deformation along the cutting edge, for position 70

6. SUMMARY

In this thesis, the stresses and deformations developed in the cutting tool during the Power skiving process were studied. The design of the cutting tool, the chips, and the analysis of the machining for nine different cutting positions were carried out using the ANSYS finite element software. Performing the analysis required the meshing process, which varied per simulation case, as well as the definition of loads and boundary conditions.

As can be seen from the diagrams, the maximum stress and deformation occur at cutting position 70, with the maximum stress reaching 6725.1 N/mm^2 and the maximum deformation reaching 0.1975 mm. Regarding the maximum stresses, it is observed that they vary depending on the chip geometry in each case. In contrast, deformation shows an almost linear increase as the cut progresses to a greater depth.

7. REFERENCES

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